SAS® Retrieval Agent Manager: User's Guide

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SAS® Retrieval Agent Manager: User's Guide

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Introduction to SAS Retrieval Agent Manager

What Is SAS Retrieval Agent Manager

Product Overview

SAS Retrieval Agent Manager is an enterprise Agentic AI solution that enables organizations to build, orchestrate, and manage intelligent agents capable of retrieval, reasoning, and task automation. By connecting enterprise data with coordinated agent workflows, it delivers context-aware responses, actionable insights, and streamlined business operations. An intuitive no-code experience allows users to quickly create and operate trusted agentic workflows, while developers maintain the flexibility to build custom agents, tools, templates, and integrations for advanced enterprise use cases.

Key Features

Rapid Agent Development with Human-in-the-Loop Controls

Reusable templates enable faster AI agent development to support rapid experimentation, accelerate iteration, and ensure human oversight with enterprise-grade compliance.

Orchestrated Multi-Agent Intelligence

Multi-agent orchestration enables complex queries to be decomposed and delegated across specialized agents, improving scalability, reasoning depth, and response quality.

Secure MCP-Based Enterprise Tool Integration

MCP tool integration enables agents to securely interact with enterprise systems through remote and custom MCP servers. It supports structured data retrieval, workflow execution, and real-world actions with robust security and secret management controls.

Built-in RAG for Proprietary Knowledge Unlocking

The RAG pipeline unlocks insights from proprietary unstructured data using multilingual OCR, optimized chunking strategies, embedding model integration, and scalable vector storage.

Agentic Retrieval Optimization

Agentic retrieval optimization enables dynamic retrieval of only the most relevant RAG chunks to improve performance, reduce cost, and accelerate response times while supporting source tagging for higher precision and contextual accuracy.

Trustworthy and Transparent AI

Robust governance controls, enabled by built-in evaluation tools and advanced observability, strengthen trust and transparency across AI workflows. Citation-backed responses, execution tracing, and comprehensive logging provide end-to-end visibility into retrieval processes, LLM interactions, and tool usage.

High Performance and Enterprise-Ready Integration

Fast, efficient AI is seamlessly integrated into enterprise environments through secure, scalable APIs and production-ready deployment infrastructure.

Flexible GenAI Stack with BYO Model Support

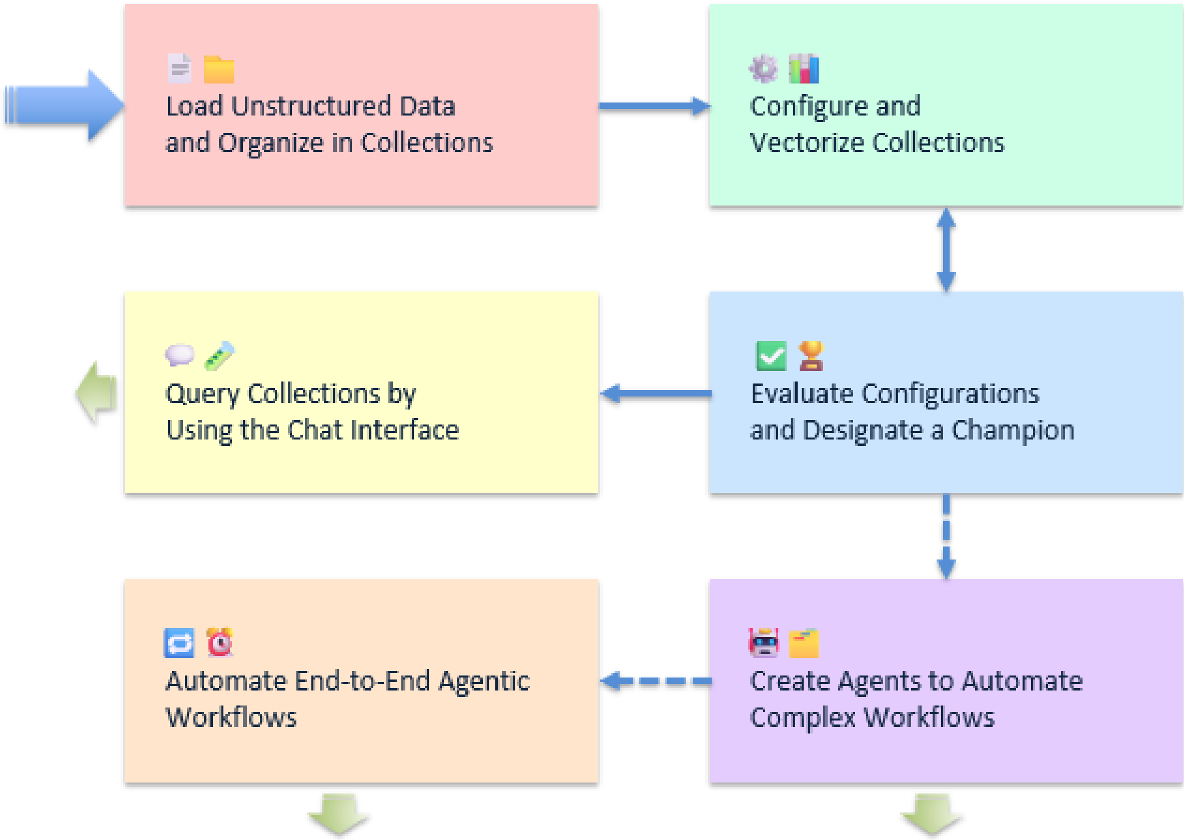
The bring-your-own (BYO) generative AI integration enables organizations to use preferred LLMs, embedding models, and vector databases while preserving architectural flexibility and vendor independence.

Enterprise-Grade Deployment Flexibility

Support for both on-premises and cloud environments aligns with organizational requirements for security, compliance, scalability, and operational control.

What Is a Typical Workflow for SAS Retrieval Agent Manager

The following figure and discussion demonstrate a typical workflow for SAS Retrieval Agent Manager.



Load unstructured data and organize into collections.

1 Upload supported file types, such as PDF, JSON, image, text, or Word.

   The file types that are supported are listed in "Supported File Types" on page 121.

2 Create and manage collections in order to logically group related documents.

For more information about these tasks, see Chapter 4, "Working with Sources," on page 19.

Configure and vectorize collections.

1 Define configurations by using embedding models, chunking strategies, and optical character recognition for scanned or image-based content.

   Note: For information about the supported embedding models, see "Overview of Integrating Content with an LLM" on page 5.

2 Initiate vectorization, and index embeddings for retrieval.

For more information about these tasks, see Chapter 6, "Working with Collections," on page 39.

Evaluate configurations and designate a champion.

1 Run evaluations to measure configuration performance and response quality.

2 Compare results across test scenarios or language models.

3 Select and designate a champion configuration for production use.

For more information about these tasks, see Chapter 6, "Working with Collections," on page 39, Chapter 7, "Working with User Evaluation Tests," on page 67, and Chapter 5, "Working with Retrieval Settings," on page 31.

Query collections by using the chat interface.

1 Interact with unstructured data by using the chat UI or API.

2 Test and iterate on chat responses to improve accuracy and relevance.

For more information about these tasks, see Chapter 12, "Working with the Chat Interface," on page 113.

Create agents to automate complex workflows.

1 Develop agents by using Python APIs to query collections and process responses.

2 Automate tasks, such as post-processing, tool invocation, and API integrations.

3 Monitor agent performance, logs, and health metrics in real-time.

For more information about these tasks, see Chapter 10, "Working with Agents," on page 93.

Automate end-to-end agent workflows.

1 Build visual pipelines that connect sources, collections, and agents.

2 Configure workflow logic, vectorization settings, and execution schedules.

3 Enable full automation of data ingestion, retrieval, and response workflows.

For more information about these tasks, see Chapter 11, "Automating Workflows," on page 107.

Overview of Integrating Content with an LLM

The following steps describe the process for integrating content with a large language model (LLM) for downstream tasks, such as submitting queries or working with agents. This is the typical retrieval-augmented generation (RAG) process.

These steps occur during the vectorization job.

1 Parsing

Parsing enables you to extract and parse text, images, and tables from different file formats. (For a list of the supported file types, see "Supported File Types" on page 121.) Documents are parsed and chunked to create text chunks.

Document parsing settings include options for maximum chunk size, optical character recognition, text cleaning, and table detection.

2 Embedding

Embedding converts parsed text into numerical vectors that capture semantic meaning for similarity search. Various embedding models enable you to generate vector representations with options to quantize and compress for efficiency.

The following embedding models are supported:

- all-Mini-LM-l6-v2
- distiluse-base-multilingual-cased-v2
- ibm-granite / granite-embedding-30m-english
- ibm-granite / granite-embedding-107m-multilingual
- ibm-granite / granite-embedding-125m-english
- ibm-granite / granite-embedding-278m-multilingual
- nomic-embed-text-v2-moe
- prdev/mini-gte

3 Indexing

Indexing stores embeddings in a structured way to enable fast and relevant retrieval. The application supports vector databases, such as Weaviate and Postgres, for indexing embeddings along with other relevant metadata. Storing the embeddings with relevant metadata enables you to apply advanced filtering at query time.

These steps occur during chat queries.

1 Retrieval

At query time, your query is used to retrieve the most relevant chunks in your documents. This is done by embedding the query with the same embedding model as in the embedding step, and comparing the query's embedding against the chunk embeddings. Typically, the Top K most relevant chunks are retrieved to include in the augmentation step.

2 Augmentation

After retrieving the most relevant document chunks from the vector database, their text content is incorporated into the original query. You can control how the chunks are augmented with your query by editing the system prompt in the Retrieval Settings view.

3 Generation

The final step in the process is for the LLM to generate a response to your query that is augmented with the most relevant document chunk text. Along with the final response, SAS Retrieval Agent Manager displays the document chunks that were retrieved for your query.

Overview of Evaluations in SAS Retrieval Agent Manager

Evaluation is a critical step in the retrieval-augmented generation (RAG) pipeline to ensure that the system produces accurate, relevant, and efficient responses. The evaluation process in SAS Retrieval Agent Manager involves measuring performance by using metrics for precision, recall, and latency. The process helps you to tune configurations, optimize configurations before deployment, and minimize irrelevant results by identifying areas for improvement.

The two types of evaluations in SAS Retrieval Agent Manager are user evaluations on page 7 and automated evaluations on page 7.

User Evaluations

User evaluation is a manual process for highly specific scenarios that require human judgment. It involves creating test cases with expected responses and keyword checks to compare responses against predefined standards. User evaluations are beneficial when testing edge cases, ensuring regulatory compliance, or validating model behavior in sensitive applications. Opik is integrated with the application to score outputs against expected results.

The User Eval Tests view in SAS Retrieval Agent Manager enables you to create and manage user evaluation tests to be used on the collections and agents in the system. For more information about working with user evaluations tests, see Chapter 7, “Working with User Evaluation Tests,” on page 67. In the Collections view in SAS Retrieval Agent Manager, the **User Eval** tab for a collection enables you to specify a predefined user evaluation test to manually evaluate the responses for a configured large language model (LLM). For more information about working with user evaluations for a collection, see “Managing User Evaluations” on page 61. In the Agents view, the **User Evaluation** tab for an agent enables you to specify a predefined user evaluation test to manually evaluate the responses across various agent experiments. For more information about working with user evaluations for an agent, see “Evaluate an Agent Experiment” on page 102.

Automated Evaluations

Automated evaluation is an auto-generated process for large-scale performance analysis. The process uses AI-driven techniques to assess retrieval and response quality across multiple queries. Large language models can assist by automatically generating test cases, which makes evaluation scalable and efficient. The following key metrics from the Ragas framework are integrated with the application. The retrieval quality of the metrics and their close adherence to facts make a rigorous evaluation of language-generation systems possible.

- Answer Relevance (For information about this metric, see Answer Relevance in the Ragas documentation.)
- Context Precision (For information about this metric, see Context Precision in the Ragas documentation.)
- Context Recall (For information about this metric, see Context Recall in the Ragas documentation.)
- Faithfulness (For information about this metric, see Faithfulness in the Ragas documentation.)

In the Collections view in SAS Retrieval Agent Manager, the **Auto Eval** tab for a collection enables you to use an LLM that is configured for a collection to automatically generate questions and answers from information in your sources. For more information about working with automated evaluations for a collection, see “Managing Auto Evaluations” on page 63.

Statement of Intended Use

Intended Use

SAS Retrieval Agent Manager is designed to assist business users, research analysts, and other stakeholders by streamlining the discovery of relevant information from large volumes of unstructured enterprise data and generating contextually appropriate responses. The solution aims to enhance user productivity and improve the accuracy of business insights. The Offering can be configured to connect to external tool servers, including servers implementing the Model Context Protocol (MCP), to extend its capabilities. Customers are responsible for selecting large language models, vector databases, configuration settings, and any external tool server connections that meet their specific requirements while ensuring compliance with applicable laws, regulations, and internal policies.

Limitations on Use

Although the solution helps mitigate risks such as hallucination, generative AI models use statistical predictions and can produce incomplete, inaccurate, or otherwise unexpected outputs. The Offering does not replace human judgment in reviewing, validating, or applying the generated outputs. Customers should validate all outputs before acting on them and implement appropriate review processes for their specific use cases.

External Tool Server Connections

SAS does not control, vet, or make any representations regarding the security, reliability, content, or behavior of third-party tool servers, including MCP servers. Customers are solely responsible for evaluating and approving any external tool server prior to connection, including assessing risks of data exfiltration, adversarial content injection, and unauthorized data access. Connecting to unsecure or untrusted tool servers can expose a customer's environment to data loss, data corruption, or security breaches. Customers must ensure that all tool server connections comply with their organization's security policies and applicable law. Customers' use of the Offering alone will not result in compliance with any applicable law or security standard.

Sign In to SAS Retrieval Agent Manager

1. In the address bar of your web browser, enter the URL for SAS Retrieval Agent Manager and press **Enter**.

   Note: Contact your system administrator if you need the URL for SAS Retrieval Agent Manager. General access to this application enables you to perform read-only actions. Contact your system administrator if you need other permissions.

2. On the Sign In page, enter a user ID and password. If configured, you can sign in by using Active Directory.
3. Click **Sign In**. The SAS Retrieval Agent Manager home page appears.

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General Usage

Overview

There are common features that are available within SAS web applications. This document provides general usage information about the following common features and application-specific options that are relevant to using SAS Retrieval Agent Manager:

- "Applications Menu" on page 12
- "Landmarks" in *SAS Viya Platform: General Usage Help for Web Applications*
- "Search" in *SAS Viya Platform: General Usage Help for Web Applications*
- "Tables" on page 12
- Application-specific options
  - "Application Settings" on page 13
  - "Navigation Pane" on page 12
  - "Help Center" on page 13
  - "Accessibility Settings" on page 13

Applications Menu

Click ⠿ in the application bar to access the list of applications that are available to you.

Tables

When you are viewing information within a table, use these tips to control how data is displayed:

- To sort information within a table, right-click a column heading and select **Sort → Sort (ascending)** or **Sort (descending)**. When the column is sorted in ascending order, ↑ appears beside the column heading. When the column is sorted in descending order, ↓ appears beside the column heading.
- To reverse a sort, click a column heading again.
- To clear the column sort, right-click the column heading, and select **Sort → Remove sorting**.
- To select which columns are displayed, right-click a column heading and select **Manage columns**. In the Manage Columns window, move the columns that you want to display to the **Display columns** pane. Click **OK** to save your changes.

Navigation Pane

The toggle button on the application banner enables you to change which views are enabled in the navigation pane based on your workflow. When the button is toggled to **Simplified workflow**, the following views are enabled:

- Dashboard
- Sources
- Collections
- Agents
- Chat

When the button is toggled to **Advanced workflow**, the following views are enabled:

- Dashboard
- Sources
- Collections
- User Eval Tests
- Retrieval Settings
- Agents
- Automation
- Chat
- MCP Tools
- Code Templates

Note: This button is enabled only for authorized users.

Application Settings

To access settings that are specific to using SAS Retrieval Agent Manager, click .

Here you can configure the large language models and destinations for the application.

Help Center

To access the product's documentation, click ? in the application bar.

Accessibility Settings

Click the user button in the application bar to access the accessibility settings that are available to you.

To meet the needs of some users with low vision, you can toggle on the **Dark Mode** to change the light color palette to an all-dark palette with white text.

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Using the Dashboard

Using the Dashboard

Note: This view is enabled only for authorized users. Authorized admin users can view information about queries made by other admin and non-admin users.

The information in this view is read-only.

Spotlight Tab

The **Spotlight** tab on the Dashboard includes bar charts that represent the following performance metrics.

Metric	Description
Query Count	The total number of queries processed per collection or agent.
Total Query Cost	The total cost of all queries per collection or agent. The cost is displayed in USD.
Average Query Cost	The average cost of a query per collection or agent. The cost is displayed in USD.

Metric	Description
Positive Response	The percentage of the query feedback that was positive per collection or agent.
Negative Response	The percentage of the query feedback that was negative per collection or agent.
Query Error Rate	The percentage of queries that resulted in an error per collection or agent.

You can adjust the information displayed in the graphs using the following filters.

Filter	Options
Type	Select whether to display performance metrics for collections or agents.
Show	Select the number of collections or agents for which you want to display performance metrics.
Period	Select the period of time for which you want to display performance metrics.

Trends Tab

The **Trends** tab in the Dashboard view provides four different combination charts that enable you to monitor cost and evaluate performance for a specified collection or agent over a selected period of time.

The **Total LLM Cost and Query Count** chart and the **Avg LLM Cost and Avg Query Cost** chart depict spending trends.

The **Avg LLM Response Time and Avg Query Duration** chart depicts LLM latency trends.

The **User Feedback %** chart depicts the proportion of positive, negative, and no-response feedback that an agent's or a collection's chat response receives in the Chat view. Analyze this chart to gain insight about the quality and effectiveness of LLM responses.

Queries Observability Tab

The **Queries Observability** tab in the Dashboard view provides insight into how queries are processed and how agents respond. Together, this information enables you to identify common usage patterns and evaluate response quality.

The **Queries Observability** tab displays detailed information about the individual queries and responses that were made over a specified period of time, including:

- The name of the collection or agent that was queried
- The type of query (either collection or agent)
- The query that was submitted by the user
- The system prompt that was generated by the LLM or agent based on the user query
- The response generated by the LLM or agent
- The start time and duration of the query response generation
- The cost associated with processing the query
- The user feedback for the query response

> **TIP** You can apply filters to the Entity Name, Entity Type, and Feedback columns in the **Queries Observability** table to refine which queries are listed in the table. For example, if you want to filter the table to display only information for your Financial News Agent, you would select the filter for the Entity Name column, select **Contains**, and then enter *Financial News*. If you want to display information for your Business News Agent as well as your Financial News Agent, click **OR**, select **Contains**, and then enter *Business News*.

The **Queries Observability** tab also displays a query activity log. For more information about the query activity log, see "View the Query Activity Log" on page 115.

Resource Consumption Tab

The **Resource Consumption** tab displays information about licensed capacity and resource usage in SAS Retrieval Agent Manager. This tab provides different visualizations that help you understand how disk space, CPU, memory, and data volume are consumed across agents, sources, embedding models, and related components.

At the top of the tab, the total licensed data volume and the amount currently used are displayed. Select **View details** to open a window with a table that shows data volume usage by date, including usage by source files and MCP tools. You can view usage for predefined time periods, such as the last week, month, quarter, or year, or specify a custom date range.

The **Embedded Models Allocation** doughnut chart displays the total allocated storage and the portion of the storage that is currently used versus available. The **Source Files Allocation** doughnut chart displays total allocated storage and current usage.

The table on the **Embedded Models Storage** tab shows how much disk space is used by embedding models.Switch to the bar chart view to assess storage consumption by individual embedding models.

The table on the **Source Files Storage** tab shows disk space usage for source files that are vectorized into SAS Retrieval Agent Manager. Switch to the bar chart view to assess storage consumption by source. This view enables you to compare disk usage across individual sources and identify high-consumption sources.

Together the visuals and tables provide aggregated usage data that helps you monitor system load, understand cost drivers, and plan for efficient operation.

4 Working with Sources

About Sources

Note: This view is enabled only for authorized users.

Sources provide the raw materials for vectorization and search. After indexing, source content is provided to a configured large language model for querying.

Sources View

The Sources view enables you to manage data sources for processing within the RAG pipeline. This view displays an alphabetical list of the sources that have been added to SAS Retrieval Agent Manager. You can add, modify, and remove sources, and select a source to see its details. You cannot remove a source that is actively being used.

When you create a source, select the type of source file, give it a unique name, and add one or more data files from a local source or a Git repository. Optionally, you can use the options on the **File Update Schedule** tab to manually synchronize files on demand or automatically synchronize files on a set schedule. For a list of the file types that are supported, see "Supported File Types" on page 121.

The following table describes the information that is displayed about each available source:

Table 4.1 Information about the Sources

Property	Description
Name	The name assigned to the data source (for example, *Annual Reports*).
Type	The type of data source. A data source can be one of the following: ■ **Local Files**: Files are uploaded directly from your machine. ■ **Git Repository**: Files are fetched directly from a Git repository. ■ **Custom**: Executes custom Python code to create and save files to the system.
File Update Schedule	(Applicable only for custom and Git repository source types.) The specified date, time, and frequency for when a custom file runs, in the format of a cron expression. A value of **Manual** indicates that no cron expression was created. **Note:** For more information about the format of a cron expression, see cron. Example: `0 0 1 * *`
Associated Collections	A comma-separated list of the collections that are associated with the source.
Last Updated	The date and time at which a source was last modified. Example: `May 30, 2035, 3:00:00 AM EST`

Table 4.2 Information about Files within a Specific Source

Property	Description
Document Name	The name of a file within the selected source. Example: `AnnualReport2019-2020.pdf`
Size	The amount of data in a file expressed as Mebibytes (MiB).

Property	Description
	Example: `30.55 MiB`
Tags	A list of tags appended to the file.
Last Updated	The datetime at which a source was last modified. Example: `May 30, 2035, 3:00:00 AM EST`

Considerations for Custom and Git Sources

When you create or modify a custom source, a good practice is to validate that the source's code runs without errors. You can trigger the code to run in the Sources view and open the **Job Log** to review the triggered event.

In the Sources table, you can see when a Git source was last updated. A good practice is to periodically fetch updates from the Git repository. In the Sources view,double-click the source then click the **Files** tab and click **Synchronize** to update the files for the Git source.

Supported OCR Languages

The following languages are supported by PaddleOCR:

- eng (English)
- chi_tra (Chinese Traditional)
- dan (Danish)
- deu (German [Deutsch])
- spa (Spanish)
- fra (French)
- ita (Italian)
- nld (Dutch [Nederlands])
- pol (Polish)
- por (Portuguese)
- ara (Arabic)

The following languages are supported by Tesseract:

- eng (English)
- chi_tra (Chinese Traditional)

- jpn (Japanese)
- dan (Danish)
- deu (German [Deutsch])
- spa (Spanish)
- fra (French)
- ita (Italian)
- nld (Dutch [Nederlands])
- pol (Polish)
- por (Portuguese)
- chi_sim (Chinese Simplified)
- osd (Orientation and Script Detection)
- equ (Math/Equation Detection)
- ara (Arabic)
- tur (Turkish)

Add a Source

1. Click to navigate to the Sources view.
2. Click **Local Files**, **Git Repository**, or **Custom**.
3. Enter a name and description for the source.

   Note: A best practice is to enter a unique, descriptive name. Do not include sensitive data in the name. In addition, the name should not contain version or environment information. A best practice for the description is to briefly summarize information about the source such as when and why someone would use it and any limitations or assumptions.

4. Complete the configuration based on the source type that you selected.

Local Files source type

1. Click **Browse Files** then select one or more files from your machine.

   IMPORTANT The following limitations apply when you upload a ZIP package:

   - The directory structure must be flat. Nested directories are not supported.
   - All files must be UTF-8 encoded.

- Files must be less than 1 MiB in size.

Git Repository source type

1 Enter the Git repository's URL then click **Test connection**.

If the connection is successful, all files from the Git repository are uploaded into SAS Retrieval Agent Manager.

2 For **File update schedule**, select whether you want to synchronize the files in SAS Retrieval Agent Manager with the files in the Git repository manually on demand or automatically on a set schedule.

3 If you select **Schedule**, use the list boxes to enter the frequency with which you want to synchronize the files.

> **Note:** When you select **Schedule**, the **Cron Expression** field automatically updates based on the schedule that you enter using the list boxes. For more information about the format of a cron expression, see cron.

Custom source type

1 Select the custom source template and template version that you want to use to create the source.

2 On the **Large Language Model** tab, for each LLM, click the box in the Retrieval Settings column and select the retrieval setting.

3 On the **Environment Variables** tab, for each environment variable for which you do not want to use the default value, click a box in the **Value** column and enter a value.

4 On the **File Update Schedule** tab, select whether you want to run your custom source code to synchronize its files manually on demand or automatically on a set schedule.

5 If you selected **Schedule**, use the list boxes to enter the frequency with which you want to synchronize the files.

> **Note:** When you select **Schedule**, the **Cron Expression** field automatically updates based on the schedule that you enter using the list boxes. For more information about the format of a cron expression, see cron.

5 Click **Save**.

The source is added to the **Sources** table.

6 (Optional) If you added a custom source, a best practice is to test that the source code runs without errors. To do so, select the source from the **Sources** table and click in the menu above the list of files for that source. Click **Trigger** to verify that you want to run the code. If the code does not run

correctly, modify the code as described in To modify a custom source type on page 28.

Tag a Source

After adding a source, you can categorize the files within the source by applying tags. Tags can then be added to queries in the Chat view to filter retrieval so that the query response includes only information from sources that have the selected tags. Tags can also be added when hardcoding a query in the code for an agent template to filter retrieval so that the query response includes only information from sources that have the selected tags.

The following example code sends the query *what is ram?* to the agent through the RAM API and instructs the agent to retrieve information from any sources that have the *ram* tag when generating its query response.

```
from typing import Any, Dict

from langchain_core.messages import HumanMessage
from sasram.agent import Client


async def exec(text: str, ram_client: Client, options[str, Any]):

ram_client.post_query(text="what is ram?", search_kwargs={"tags":
["ram"]})


agent = ram_client.get_langgraph_agent()

session_history = ram_client.get_session_history_as_langchain()
messages = session_history + [HumanMessage(context=text)]

response = await agent.ainvoke({"messages": messages})
return response["messages"][-1].content
```

By limiting retrieval to the most relevant sources, tags help produce more accurate and contextually appropriate query responses. This targeted retrieval also reduces the amount of data the agent or LLM must process, which can improve response time and lower compute cost when generating answers.

To add a tag to a source:

1. Click to navigate to the Sources view.
2. In the table that lists existing sources, double-click the source that contains the file that you want to tag.
3. Click the **Files** tab then select the file in the table.

4. If you want to add a new tag to the list of available tags, click Add new tag, then enter a unique name and an informative description for the tag. Next, select whether the tag is global (available for all sources) or local (available only for this specific source) then click **OK**.

   The tag is displayed in either the list of local tags or the list of global tags depending on your selection in Step 3 on page 24.

5. To add a tag to the file, click **+** next to the tag that you want to add.

6. Revectorize the collection to make the tag available in the list of tags in the Chat view. For more information about revectorizing a collection, see “Run a Vectorization Job Manually” on page 57.

Remove Tags from a Source

1. Click to navigate to the Sources view.
2. Select the source that contains the file from which you want to remove tags.
3. Click the **Files** tab.
4. In the table, select file from which you want to remove tags.
5. Under **Tags**, click **X** next to the tag name to remove it from the file.

Add Custom Metadata Columns to a Source

If agentic retrieval is enabled and custom metadata columns are applied to a source in a collection, when you chat with an agent that is associated with that collection, the agent can combine semantic search with filters on user-defined metadata. This improved retrieval process results in accurate, relevant responses without relying only on embeddings.

1. Click to navigate to the Sources view.
2. Select the source that contains the file to which you want to apply custom metadata columns.
3. Click the **Files** tab.

4 In the table, select the .csv or .xls file to which you want to apply custom metadata columns.

> **IMPORTANT** The .csv or .xls file must contain column headings.

5 Click **Add columns**.

6 On the Add metadata column window, enter a name for the custom metadata column.

> **IMPORTANT** The name of the metadata column must match exactly the name of a column heading in the .csv or .xls file.

7 Click **Add**.

8 Click **Save**.

9 Manually run a vectorization job for the champion configuration for each collection that includes the source. For step-by-step instructions, see “Run a Vectorization Job Manually” on page 57.

Remove Custom Metadata Columns from a Source

1 Click 🗋 to navigate to the Sources view.

2 Select the source that contains the file from which you want to remove custom metadata columns.

3 Click the **Files** tab.

4 In the table, select the .csv or .xls file from which you want to remove custom metadata columns.

5 Under **Custom metadata columns**, click **X** next to the custom metadata column name to remove it from the file.

View Existing Sources and Their Details

To see a list of the sources that have been created, click to navigate to the Sources view. The existing sources are listed in the Sources table. By default, items in this view are sorted alphabetically by the Name column. You can sort the list based on values in any column or enter a string in the search field to find a specific item. The properties that are shown in this view are described in "About Sources" on page 19.

To see the files that are associated with a specific source, double-click the name of a source in the **Sources** table then click the **Files** tab.

For sources that belong to the local files source type, you can select a file name and click to open its contents in a preview window. To exit the preview window, click **Close**. To download the file, select the file name and click to save a local copy.

Note: The specific behavior for saving a local copy depends on your browser.

Manually Run a Custom Source Job

When you create a custom source as described in "Add a Source" on page 22 and do not schedule the job code to run automatically, the custom code does not run unless you manually trigger the job execution. Here is how to trigger a custom source to run:

1. Click to navigate to the Sources view.
2. Select the name of the custom type source that you want to run. You can select only a source that is designated with Manual in the Schedule column.
3. Select the **Jobs** tab below the Sources table.
4. Click in the menu above the list of files for that source.
5. Click **Trigger** to verify that you want to trigger the custom source.
6. (Optional) To verify that the job ran successfully, click . Click **Close** to exit the log window.

If the code does not run correctly, modify the code as described in To modify a custom source type on page 28.

Modify an Existing Source

Depending on the source type, you can modify certain properties such as the name and schedule, and add, remove, or update files that are associated with a source.

1 Click to navigate to the Sources view.

2 Select the name of the source that you want to modify.

To modify a local source type:

1 Double-click the source's name in the **Sources** table.

On the **Details** tab, edit the source's name and description as needed.

2 On the **Files** tab, upload and delete files in the source as needed.

- To upload a file, click then select one or more files and click **Open**.
- To delete a file, select the file in the table, and then click .

3 Add tags to the files as described in "Tag a Source" on page 24.

4 Add custom metadata columns to .csv and .xls files as described in "Add Custom Metadata Columns to a Source" on page 25.

5 Click **Save**.

To modify a Git source type:

1 Double-click the source's name in the **Sources** table.

2 On the **Details** tab, edit the source's name and description as needed. If necessary, update the Git repository URL and test the connection.

3 On the **Files** tab, add tags to the files as described in "Tag a Source" on page 24.

4 Add custom metadata columns to .csv and .xls files as described in "Add Custom Metadata Columns to a Source" on page 25.

5 To fetch updates to files in the Git repository that is associated with the source, click **Synchronize** above the source's list of files.

6 On the **File Update Schedule** tab, update the schedule as described in Step 3 on page 23.

7 Click **Save**.

To modify a custom source type:

1 Double-click the source's name in the **Sources** table.

2 On the **Details** tab, edit the source's name and description as needed. If necessary, select a different template and specify the version of the template that you want to use.

3 On the **Files** tab, add tags to the files as described in "Tag a Source" on page 24.

4 Add custom metadata columns to .csv and .xls files as described in "Add Custom Metadata Columns to a Source" on page 25.

5 On the **Large Language Models** tab, select different retrieval settings for the LLMs as needed.

6 On the **Environment Variables** tab, edit values for environment variables as needed. If necessary, use the secret column to mark values that contain confidential information.

7 On the **File Update Schedule** tab, update the schedule as described in Step 5 on page 23.

8 Click **Save**.

9 (Optional) Test that the source code runs without errors as described in "Manually Run a Custom Source Job" on page 27.

Delete a Source

> **IMPORTANT** You cannot delete a source that is actively being used in a collection. Before you can delete the source, you must delete the collection. For steps to delete a collection, see "Delete a Collection" on page 61.

1 Click to navigate to the Sources view.

2 Click the name of the source that you want to delete.

3 Click .

4 Click **OK** to confirm deletion.

The source is removed from the Sources table.

5

Working with Retrieval Settings

About Retrieval Settings

Note: This view is enabled only for authorized users.

Retrieval Settings View

The Retrieval Settings view enables you to configure parameters that are related to how the system retrieves context from your documents and how large language models (LLMs) use this to answer queries. Settings include the system prompt, the query result limit (Top K), and any additional retrieval parameters to improve response accuracy and efficiency.

In the Retrieval Settings view, you can see a list of the retrieval settings that are configured for the system. The application includes the following retrieval settings with preset system prompts:

Retrieval Setting	System Prompt
Concise	Use the given context to concisely answer the query. If you do not know the answer, say you do not know. Context: {context}
Context	Use the given context to answer the question. If the answer is not in context, say you do not know. Context: {context}
Default	The context provided is: {context}. Generate a response that not only answers the query directly but also provides relevant context, background, and related insights to enhance the understanding of the topic.
Detailed	The user is seeking assistance based on the following context: {context}. Use the context to propose practical recommendations, solutions, or best practices that are well-supported by the data.
Many Chunks	The context provided is: {context}. Generate a response that not only answers the query directly but also provides relevant context, background, and related insights to enhance the understanding of the topic.
Summary	Given the context: {context}, generate a clear summary that directly answers the query. Ensure that the response is factual, accurate, and supported by the retrieved documents.

You can add, modify, and delete retrieval settings. Retrieval settings must be mapped back to relevant collections as described in "Configure LLMs to Use within a Collection" on page 56.

The following table describes the information that is displayed about each available retrieval setting:

Table 5.1 Information about the Sources

Property	Description
Name	The name of the retrieval setting. Example: `Summary`
System prompt template	The instructions that are given to the LLM. This prompt controls how the LLM behaves. You could use this prompt to give the LLM a persona or give it explicit instructions for interacting with users, for example. **IMPORTANT** You must include the context variable `Context:` *{context}* in your text string. The {context} variable is replaced with your document content.

Property	Description
	Example: `Use the given context to concisely answer the query. If you do not know the answer, say you do not know. Context: {context}` See “Example of a Prompt with Context” on page 34.
Query result limit	(Optional) The amount of context that is returned from the retrieval process. The amount of context is measured in either chunks or tokens. For example, if you specify *10* chunks, then the top 10 chunks from the vector database that are the most relevant to a query are given to the LLM as part of the input. By default, the value for this field is 10 chunks. If you want to ensure the LLM always receives the same amount of context regardless of chunk size, use tokens instead. When using tokens, the best practice is to set the limit to at least 5000 tokens. Example: `10`
Score threshold	(Optional) A toggle for whether to filter query results based on their similarity scores. Chunks with a score below the specified threshold are ignored. The value must be between 0 and 1. Example: `0.8`
Retrieval type	The method for retrieving context from your source documents. The amount of context that you can retrieve is limited based on the number of chunks or tokens that you entered in the **Query result limit** field. Standard retrieval type is the default option. This retrieval type provides the LLM with the most relevant context. Title Chunk retrieval type is recommended only for highly structured documents where sections are organized with headings or titles. This retrieval type provides the LLM with the most relevant context in addition to all other context that is included within the corresponding section of the document. Adjacent Chunk retrieval type provides the LLM with the most relevant context in addition to the chunks of context that are located immediately adjacent to them.
Table retrieval settings	Settings that enable you to determine how context chunks are retrieved from tables that are included in your source documents. You can configure the maximum tokens retrieved for each table and determine whether to include adjacent table chunks. The recommended maximum number of tokens is 5000.
Extra retrieval parameters	(Optional) An advanced property for specifying destination-specific parameters. **Note:** Consult the documentation for your vector database for recommended parameters. Example: `alpha: 0.5` as defined in Search mechanism for Weaviate.

When you create or modify a retrieval setting, the **Template** field in the Add Retrieval Settings and Edit Retrieval Settings windows is the **System prompt** that is displayed in the Retrieval Settings view.

Example of a Prompt with Context

In this example of a prompt with context, you ask the question, "What types of windows are available in ESP?" This query uses the **Concise** retrieval setting and the **Query result limit** is set to `5`. Here is the prompt to the LLM:

```
Use the given context to concisely answer the query. If you do not
know the answer, say you do not know. Context:
  1.'Window Types - An event stream processing project specifies how
input event streams are transformed and analyzed into meaningful
resultant event streams. Each project contains one or more continuous
queries...'
  2.'Counter window - Enables you to see how many events are streaming
through your model and the rate at which they are being processed...'
  3.'Functional window - Enables you to use different types of
functions to manipulate or transform the data in events. Fields in a
Functional window can be hierarchical, which can be useful for
applications such as web analytics...'
  4.'Geofence window - Enables you to create a window to determine
whether the location of an event stream is inside or near an area of
interest...
  5.'Join window - Takes two input windows and a join type. Supports
equijoins that are one to many, many to one, or many...
```

Here is the answer that is returned:

```
Event stream processing (ESP) has several different window types,
including counter windows, functional windows, geofence windows, and
join windows.
```

Add Retrieval Settings

1 Click to navigate to the Retrieval Settings view.

2 Click .

3 In the Add Retrieval Settings window, add values for the following properties:

Note: The properties in the Add Retrieval Settings window are described in "About Retrieval Settings" on page 31.

a Enter a **Name** for the retrieval setting.

b Enter the **System prompt template**.

c (Optional) Change the default value for the **Query result limit**.

d (Optional) To filter query results based on their similarity score, toggle on the **Score threshold**. Enter a different value if you do not want to accept the default value.

e Select a retrieval type.

Note: For more information about retrieval types, see "About Retrieval Settings" on page 31.

If you select **Title chunk retrieval** as your retrieval type, enter the maximum number of chunks or tokens that you want to retrieve from the corresponding section of the document.

If you select **Adjacent chunk retrieval** as your retrieval type, enter the number of adjacent chunks that you want to retrieve.

f For **Table retrieval settings**, configure the following settings:

- **Include adjacent chunks**

  Enable this setting to provide the LLM with additional context from the chunk before and the chunk after the retrieved table.

- **Max tokens per table**

  Enter the maximum number of tokens that can be retrieved from each table in the source documents.

g Select whether to enable hybrid search settings.

TIP When you enable hybrid search settings, the LLM or agent uses a combination of a semantic meaning-based search and an exact-match keyword search to retrieve information. As a result, you might experience improved retrieval performance.

h If you enabled hybrid search settings, enter a number from zero to one.

By default, the value is `0.5`. This approach balances the utilization of exact-match keyword search functionality equally to utilization of semantic meaning-based search functionality.

Enter a number closer to zero to indicate that the LLM or agent should use exact-match keyword search functionality more than semantic meaning-based search functionality. This approach is useful for situations where you need to retrieve information from structured data based on exact identifiers or specific phrases.

Enter a number closer to one to indicate that the LLM or agent should use semantic meaning-based search functionality more than exact-match keyword search functionality. This approach is useful for situations where you need to retrieve information from unstructured data based on context clues and natural language processing.

i (Optional) To add **Extra retrieval parameters**, click + and specify the retrieval parameter.

Note: Consult the documentation for your vector database for recommended parameters.

4 Click **OK** to save your changes.

The new retrieval setting is added to the existing settings by name in alphabetical order.

IMPORTANT You must map the retrieval setting to the relevant collections as described in "Configuring the LLMs" on page 56.

Modify Retrieval Settings

1. Click to navigate to the Retrieval Settings view.
2. Click the retrieval setting name that you want to modify.
3. Click .
4. In the Edit Retrieval Settings window, you can modify the **Name** and **Template**. These properties are required. Optionally, you can modify the **Query result limit**, **Score threshold**, and **Extra retrieval parameters**. These properties are described in "About Retrieval Settings" on page 31.
5. Click **OK** to save your changes.

Delete Retrieval Settings

IMPORTANT You cannot delete a retrieval setting that is being used by an agent. Before you can delete the retrieval setting, you must assign the agent to a different retrieval setting. For steps to modify an agent, see "Modify an Agent" on page 104.

1. Click to navigate to the Retrieval Settings view.
2. Click the retrieval setting name that you want to delete.

3 Click .

4 Click **OK** to verify the action.

The retrieval setting is removed from the Retrieval Settings view.

View Existing Retrieval Settings and Their Details

To see a list of the retrieval settings that have been created for SAS Retrieval Agent Manager, click to navigate to the Retrieval Settings view. The existing retrieval settings are listed in this view. By default, items are sorted alphabetically by retrieval setting name.

To see the details that are associated with a specific retrieval setting, select the setting's name. The configurations for that setting open in the right pane.

The properties that are shown in the Retrieval Settings view are described in "About Retrieval Settings" on page 31.

6

Working with Collections

About Collections

Note: This view is enabled only for authorized users.

Creating a collection is a foundational step in the retrieval-augmented generation (RAG) pipeline. Collections allow the system to retrieve the most relevant documents based on the user's query, which improves accuracy and response quality. Without collections, responses would be less precise and contextually not relevant because they would lack the required structured knowledge. For information about a typical workflow for SAS Retrieval Agent Manager, see "What Is a Typical Workflow for SAS Retrieval Agent Manager" on page 3.

The Collections view enables you to manage and organize document collections for RAG applications. In this view, you can see existing collections and their associated sources, create, modify, and delete collections, and open a collection to manage its configurations.

Configuring a collection involves setting up parameters that define how document vectorization, retrieval, and query processing function. This includes specifying the vector database where embeddings are stored and creating a vectorization job to transform raw text from the source documents into high-dimensional numerical vectors. These vectors enable efficient similarity-based searching, ensuring that the most relevant documents are retrieved. Proper configuration ensures that the system optimally indexes data, which improves performance and response accuracy. Because there are multiple ways to generate embeddings based on configurations, collections enable you to create and test different settings to optimize retrieval performance by using an evaluation framework, such as user evaluations and automated evaluations.

Collections also enable you to select and assign one or more large language models (LLMs) to use for querying and evaluation. These LLMs are also used by agents. In addition, collections can serve as a security management tool by automatically creating security groups that control user access. This ensures that only authorized users can query certain document sets. This makes collections an essential feature for structured document access and permission control.

Collections View

The Collections view displays existing collections as tiles. Each tile contains basic information, such as the collection name and its associated sources. These tiles are organized by name in alphabetical order, from left to right.

In this view, you can add a new collection, search for a specific collection, and delete a collection. You can modify the collection's properties, such as **Name** and **Description** but you cannot change a collection's associated sources.

Details for a Collection

You can select a tile on the Collections view to see the details about that collection, such as configuration details and details about user and automated evaluations. Each collection includes **Summary**, **LLMs**, **Auto Eval**, and **User Eval** tabs. A tab with details for each named configuration within a collection is also included.

Summary Tab

The **Summary** tab lists the configurations that have been created for a collection and the details about a configuration's performance. The following table lists the details that are documented for each configuration.

Table 6.1 *Details about a Collection's Configurations*

Metric	Description
Configuration	The name of the configuration. Example: `bge-base`
Vectorized Status	The status of the vectorization job the last time the configuration was vectorized. The status can be **Cancelled**, **Completed**, **Error**, **Queued**, or **Running**. A job log is also available in this column for each configuration. A log is also available in this column for each vectorization job. Example: `Completed`
Last Job	The datetime at which the configuration last ran. Example: `May 30, 2035, 3:00:00 AM EST`
Last Auto Eval	The Retrieval-Augmented Generation Assisted Score (Ragas), which indicates how well a RAG model is performing in terms of relevance, faithfulness, precision, and recall. This score updates every time you run a user evaluation test. Example: `64%`
Last User Eval	A link for the number of user evaluation prompts for a configuration that passed during the last user evaluation job. The score is based on the most recent evaluation run, not historical averages. **Note:** You can click the link in this column to see information about the prompts that passed or failed for the configuration. Example: `1/2`
Configuration Status	A designation for the configuration. A configuration can be **Under Evaluation**, a **Challenger**, or a **Champion**. **Note:** A configuration that is Under Evaluation is a new configuration that has not undergone testing. A Challenger designation means that the configuration is being tested. A Challenger can be used in automated pipelines. (See "About Automation" on page 107). A Champion configuration is the configuration designated for use within the application for

Metric	Description
	querying. Only the Champion configurations can be used by agents and end users. Example: `Champion`

When you select a configuration in the summary table, the configuration's details are listed below the summary table in the **Basics**, **Settings**, and **Text Extraction** sections. These sections are tabs in the New Configuration window. The information about these tabs is described in "New Configuration Window" on page 48.

LLMs Tab

The **LLMs** tab enables you to configure large language models (LLMs) for use within a collection or for evaluations. The left pane on the tab displays all LLMs that are available in the system. Icons next to each model indicate the model provider, such as Microsoft Azure, an OpenAI endpoint, or a local model. ✓ for an available LLM means that the connection is active and healthy. ⃠ for an available LLM means the connection failed due to an error.

The top right pane on the tab shows the LLMs that are currently assigned to the collection. The bottom right pane shows the LLMs that are configured for **User Eval** and **Auto Eval**. You can reorder the list of configured LLMs. The order reflects the LLM priority that is seen by users on the Chat view. In that view, users without administrative privileges can see only the first model in the ordered list. Users with administrative privileges can see all configured models.

Configuring LLMs for a collection enables you to interact with the chat on page 113 user interface and run user evaluations on page 62.

Auto Eval Tab

Note: For an overview of automated evaluation, see "Overview of Evaluations in SAS Retrieval Agent Manager" on page 7.

The **Auto Eval** tab enables you to use an LLM that is configured for a collection to automatically generate questions and answers by using information from your source documents. In this process, you tell the system to randomly select a predefined number of chunks, give the LLM the chunk, and ask the LLM to create a question that would require using that chunk for the answer.

The Auto Eval Data table at the top of the tab lists information about the evaluation data and the questions that have been generated for the specific collection. In this section, you can generate new evaluation questions, modify an existing question, and delete a question. When you generate new evaluation questions, you specify the following:

Table 6.2 Specified Properties to Generate Evaluation Data

Property	Description
Source coverage (%)	The percentage of content that should generate a question. Example: `10`
Num test cases to be generated	The number of test cases that should be generated for the evaluation. This value changes automatically depending on the value for Source coverage (%). Example: `8`
Large language model	The LLM that is used for the evaluation job. Example: `gpt-4o`
Reference configuration	The configuration that is tested with the evaluation job. Example: `gte-base-wv`

The following table describes the information that is displayed in the Auto Eval Data table:

Table 6.3 Details about Automated Evaluation Data

Property	Description
Question	A generated question that is used to evaluate the RAG pipeline. Example: `Compare and analyze the numerical values given in the three sets of data. What patterns or trends can you observe from the comparisons?`
Context	The context (chunk of data) that was used to generate the question. Example: `14.0 vs. 10.0 13.5 vs. 11.3`
Ground Truth	The reference answer that is used to evaluate the RAG pipeline. Example: `The three sets of data provided each consist of two numerical values being compared. In each set, the first value is...`
Generator LLM	The LLM that generated the reference answer and question. Example: `gpt-4o`
Cost	The LLM cost of generating the question and ground truth data in USD.

Property	Description
	Example: `$0.1612`

The Auto Eval Results table at the bottom of the **Auto Eval** tab lists the automated evaluation jobs and metrics about each run. In this section, you can filter out stale results, run an automated evaluation job, delete items from the results table, and view the logs from the jobs. The following table describes the metrics that are displayed in the Auto Eval Results table:

Table 6.4 Details about Automated Evaluation Metrics

Property	Description
Timestamp	The datetime at which the evaluation last ran. Example: `May 30, 2035, 3:00:00 AM EST`
Configuration	The configuration that was used for the evaluation job. Example: `gte-base`
Eval LLM	The LLM that was used to generate the answers for the evaluation job. Example: `gpt-4o`
Critic LLM	The LLM that was used to evaluate the outputs for the evaluation job. Example: `gpt-4o`
Critic Embedding Model	The embedding model that was used to evaluate the outputs for the evaluation job. **Note:** For information about the supported embedding models, see "Overview of Integrating Content with an LLM" on page 5. Example: `gte-base`
Status	Status of the job run. The status can be **Cancelled**, **Completed**, **Error**, **Queued**, or **Running**. A log is also available in this column for each evaluation job. Example: `Completed`
Ragas Score	The summary score of all metrics for the job. The Ragas score is the harmonic mean of faithfulness, answer relevancy, context recall, and context precision. This score updates every time you run a user evaluation test. Example: `64%`
Cost	Cost of the job in USD. Example: `$4.58104`

Property	Description
Details	A link that opens the Evaluation Results window, which provides the detailed metrics and outputs for each question for the job. The following table described the information in this table:

Metric	Description
Question	A generated question that is used to evaluate the RAG pipeline.
Ground Truth	The reference answer that is used to evaluate the RAG pipeline.
Reference Contexts	The context (chunk of data) that was used to generate the question.
Response	The generated answer from the RAG pipeline for the question.
Retrieved Contexts	The chunks that were retrieved by the RAG pipeline for the question.
Ragas Score	Individual computed Ragas score for the question.
Faithfulness	A measurement from 0 to 1 for how factually consistent a response is with the retrieved context. A higher score indicates better consistency. For more information about this metric, see Faithfulness in the Ragas documentation.
Answer Relevancy	A percentage for how pertinent the generated answer is to the given prompt. For more information, see Answer Relevance in the Ragas documentation.
Context Recall	A percentage for how many of the relevant documents (or pieces of information) were successfully retrieved. For more information, see Context Recall in the Ragas documentation.

Property	Description	
	Context Precision	A percentage for the proportion of relevant chunks in the retrieved_contexts. For more information, see Context Precision in the Ragas documentation.
	Cost	The LLM cost of evaluating the question.

User Eval Tab

Note: For an overview of user evaluation, see "Overview of Evaluations in SAS Retrieval Agent Manager" on page 7.

The **User Eval** tab enables you to manually evaluate the responses for an LLM by using predefined test cases. You can do the following tasks on this tab:

- See details about previous evaluation runs.
- Run new evaluation jobs for specific configurations.
- View the exact prompts, responses, assertions, and pass or fail results for a specific run.
- Delete a run from the evaluations table.

IMPORTANT Before you can run an evaluation job, you must first create evaluation test cases. For more information about evaluation test cases, see "About User Evaluation Tests" on page 67.

The following table describes the details that are documented for each evaluation job that is run for a configuration:

Table 6.5 Details about User Evaluation Jobs

Property	Description
Timestamp	The datetime at which the evaluation last ran. Example: `May 30, 2035, 3:00:00 AM EST`
Test Case	The name of the test case that was used for the evaluation job. Example: `Financial questions`

Property	Description
Eval LLM	The LLM that was used to generate the answers for the evaluation job. Example:
Configuration	The configuration that was tested with the evaluation job. Example: `gte-base`
Status	The status of the evaluation job. The status can be **Cancelled**, **Completed**, **Error**, **Queued**, or **Running**. A log is also available in this column for each evaluation job. Example: `Completed`
Score	The number of user test cases that passed during the evaluation job. The score is based on the most recent evaluation run, not historical averages. Example: `1/2`
Cost	The LLM cost of evaluating the question in USD. Example: `$0.016270`

When you select a specific job from the evaluations table, the following information about the run is displayed at the bottom of the **User Eval** tab:

- A list of the **User eval prompts** that are associate with the test case for the selected evaluation job.
- The full text of a selected **Prompt**.
- The **Response** to the prompt.
- The **Assertions** that were made for the prompt, which include a **Metric** (for example, `contains`), the **Reference** (for example, `csv`), and the **Result** (for example, `Pass`).
- The **Passing threshold** that specified the percentage of assertions that have to pass for the prompt.

Named Configurations Tabs

Each collection contains a tab for every configuration that is created for the collection. The **Details** section for a configuration tab displays the settings that were specified when the configuration was created. These settings are Read-only and cannot be modified. They are described in "New Configuration Window" on page 48.

The **Documents** section shows all the documents that are associated with the source collection. You can click **Details** in a document to view the chunks of content. By default, all chunks can be retrieved during a search or query. You can toggle **Enable retrieval** off for a specific chunk that should not be retrieved during a search or query.

New Configuration Window

You can create one or more configurations for a collection. All configurations within a collection share the same source documents that are associated with the collection. In the New Configuration window, you specify the configuration's settings on the **Basics**, **Settings**, and **Text Extraction** tabs.

> **IMPORTANT** You cannot modify a configuration's settings once you save it. You can copy the configuration to create a new configuration or you can delete the configuration from the collection.

Basics Tab

The following table describes the configuration fields on the **Basics** tab.

Table 6.6 Specifications for a New Configuration on the Basics Tab

Field	Description
Name	The name of the configuration. Example: `bge-base`
Description (Optional)	A string that describes the configuration.
Embedding model	A model that converts text into vector embeddings. The model can be **all-MiniLM-L6-v2**, **bge-base-en-v1.5**, **bge-small-en-v1.5**, **gte-base**, **gte-small**, or **multilingual-e5-small**. Example: `all-MiniLM-L6-v2`
Enable compression (Optional)	A toggle for whether to enable compression.
Enable quantization (Optional)	A toggle for whether to enable quantization.
Destination	The target vector database destination. The destinations that are available depend on what is configured for your system. The

Field	Description
	destination can be **pgvector storage default**, **pgvector VHUB Test CMD**, **weaviate storage cloud**, or **weaviate storage local test**. Example: `pgvector storage default`
Configuration update strategy	A strategy for what happens when the source documents are updated and indexed. The available strategies are described here. Depending on your choice, the previous embeddings could be retained or replaced. ■ **Disabled**: No synchronization is done between the source documents and the vector database. This means that for each vectorization job, all source document embeddings are added to the vector database, even if they already exist from a previous vectorization job. ■ **Append**: Only new embeddings (for example, from a new or updated source document) are added to the vector database. Existing embeddings already present in the vector database are skipped. No embeddings are deleted from the vector database, even if the original source document is no longer present. ■ **Append and sync**: This strategy is the same as the **Append** strategy, but when a source document is updated, any outdated embeddings are also deleted from the vector database. ■ **Append, sync, and delete**: This strategy is the same as the **Append and sync** strategy, but it also removes embeddings from documents that are no longer in the source. In this scenario, only embeddings from the latest source documents exist in the vector database. Use this strategy if you want to ensure that your vector database always contains exactly one copy of embeddings for the current source documents. Example: `Append` For more information about how to configure an update strategy, see “Example for Configuration Update Strategy” on page 49.

Example for Configuration Update Strategy

Source A contains two documents, Document A and Document B, and is used in Collection A.

1 Create the following configurations:

- Configuration A uses the **Disabled** strategy.
- Configuration B uses the **Append** strategy.

- Configuration C uses the **Append and sync** strategy.
- Configuration D uses the **Append, sync, and delete** strategy.

2 Execute a vectorization job for all four configurations. Two documents are embedded in each job.

3 Execute another vectorization job without changing any of the source documents. In this case, Configuration A now contains embeddings for four documents (the original two documents are duplicated). Configurations B, C, and D continue to hold embeddings for the two documents.

4 Update Document A by replacing the original content and resubmit the vectorization jobs for each configuration. Here are the results:

- Configuration A now contains the embeddings for documents from the two previous jobs and the embeddings from this most recent job, which are for the updated document and the unchanged Document B.
- Configuration B now contains the embeddings from the original Document A and the updated Document A, along with the embeddings from the unchanged Document B.
- Configurations C and D now contain the embeddings only from the updated Document A (the original Document A embeddings are deleted) and the embeddings from the unchanged Document B.

5 Delete Document B because it is no longer needed, and resubmit the vectorization jobs for each configuration. Here are the results:

- Configuration A now contains the embeddings from the previous three jobs, plus the embeddings from the latest job.
- Configurations B and C skip adding the embeddings from Document A because the embeddings are already present but do not remove the embeddings from Document B, which were added in the original job.
- Configuration D skips adding the embeddings from Document A because the document is unchanged and embeddings are already present. The embeddings from Document B are removed because Document B is no longer present in the source.

Settings Tab

The following table describes the configuration fields on the **Settings** tab.

Table 6.7 Specifications for a New Configuration on the Settings Tab

	Field	Description
Document ingestion	Exclude the following extensions from	The option to exclude file extensions, such as .csv or .pdf, from processing. Only supported file types appear if they exist in the associated sources.

	Field	Description
	processing (Optional)	
	Clean text after ingestion (Optional)	A toggle for whether to clean the text after ingestion. When enabled, you remove special characters (for example, non-ASCII, Unicode quotation marks, and bullets) from your document, which improves embedding quality.
Chunking Strategy	Chunk size (Optional)	The number of tokens per chunk of content. The default is `Max`, which is the maximum chunk size allowed by the embedding model. Example: **`Max`** (For example, **`Max`** for the all-Mini-LM-l6-v2 embedding model is 512 tokens.)
	Chunk overlap (Optional)	The overlap to add between the chunks for better semantic continuity. Example: **`0`**

Text Extraction Tab

The following table describes the configuration fields on the **Text Extraction** tab.

__Table 6.8__ Specifications for a New Configuration on the Text Extraction Tab

Field		Description
Enable OCR (Optional)		A toggle for whether to enable optical character recognition for scanned documents or images within text.
OCR settings	OCR library	The library to use for optical character recognition. You can select **PaddleOCR** or **Tesseract**. Example:
	OCR language (Optional)	The language to use for optical character recognition. Example: **`ENG`**
Export tables (Optional)		A toggle for whether to represent tables in your documents as structured strings (for

Field		Description
		example, markdown) for storage in a vectored database.
Table extraction settings	Table detection pipeline	The pipeline used for detecting tables. Values are **Default** and **Lightweight**.
	Table export format	The format that you select for exported tables. Values are **HTML** and **Markdown**. Example: `Markdown`
	Implicit row detection (Optional)	A toggle for whether to detect implicit rows in unstructured tables.
	Implicit column detection (Optional)	A toggle for whether to detect implicit columns in unstructured tables.
	Detect borderless tables (Optional)	A toggle for whether to detect borderless tables.
	Table detection threshold (0-100) (Optional)	The confidence threshold for detected tables. Tables are detected only by the table detection model if their confidence value is above this threshold. Example: `50`

About Vectorization Jobs

After you create a configuration, you start a vectorization job to ensure that you can retrieve relevant chunks of data for the target vector database. You can specify the following settings for vectorization:

Note: Later, you can set a vectorization job to run automatically. For more information about automation, see "About Automation" on page 107.

Table 6.9 Settings for Running Vectorization Jobs

Setting	Description
CPU	The number of CPUs that are available for vectorization. Example: `6`

Setting	Description
Memory	The amount of memory that is available for vectorization. Example: `4Gi`
Execution provider	The execution resource. Values are **CPU** or **OpenVINO**. **Note:** Although both execution resources produce the same results, vectorization jobs might run faster when you select **OpenVINO**. Example: `CPU`
OpenVINO device	(Enabled if **Execution provider** is **OpenVIno**) The device that runs the OpenVINO resource. Example: `CPU_FP32`

Creating a Collection

Overview of Creating a Collection

The properties that you can specify to create a collection are described in "Details for a Collection" on page 40.

Creating a new collection involves these steps:

- Define a collection's properties. on page 53
- Create a configuration. on page 54
- Manually run a vectorization job. on page 57
- Configure the LLMs. on page 56

Define a Collection's Properties

1. Click to navigate to the Collections view.
2. Click the **New Collection** tile.
3. Enter a descriptive **Name** for the collection. The collection name should be unique.

4 (Optional) Enter a **Description** for the new collection.

5 Select one or more **Sources** and click +›.

6 Click **Save**.

The new collection is added to the Collections view.

Create a Configuration

IMPORTANT You cannot modify a collection's configuration settings once you save it. You can copy the configuration to duplicate its settings and then change specific settings. For more information, see "Modify a Collection's Configurations" on page 59.

The properties that you can specify when you create a configuration are described in "New Configuration Window" on page 48.

1 Click to navigate to the Collections view.

2 In the Collections view, select the collection tile for which you want to create configurations. The collection opens to the **Summary** tab.

> **TIP** To find a specific collection, begin entering the collection name in the search field.

3 Click .

4 On each tab in the New Configuration window, specify the desired settings for the configuration.

On the Basics Tab

1 Enter a **Name** for the configuration.

2 (Optional) Enter a **Description** of the configuration.

3 Select the **Embedding Model** that you want to use.

> **TIP** Smaller models use less memory. This enables you to optimize storage by using compression and quantization.

4 (Optional) If the selected embedding model supports compression or quantization, you can select the toggle to on for **Enable compression** and **Enable quantization**.

5 Select the target vector database **Destination**.

6 Select a **Configuration update strategy**. You can select either **Disabled**, **Append**, **Append and sync**, or **Append, sync, and delete**.

On the Settings Tab

1 (Optional) Expand the **Document ingestion** field.

   a (Optional) Select the file extensions that are listed if you want to exclude them from processing.

   b (Optional) To **Clean text after ingestion**, select the toggle to on.

2 (Optional) Expand the **Chunking strategy** field.

   a (Optional) Enter a value for the **Chunk size**.

   b (Optional) Enter a value for the **Chunk overlap**.

On the Text Extraction Tab

1 (Optional) To **Enable OCR** , select the toggle to on.

   a Select the **OCR library** to use. You can select **PaddleOCR** or **Tesseract**.

   b (Optional) Select the **OCR language** to use.

   Note: For more information, see "Supported OCR Languages" on page 21.

2 (Optional) To **Export tables**, select the toggle to on.

   a Select the **Table detection pipeline**. You can select **Default** or **Lightweight**.

   b Select the **Table export format**. You can select **HTML** or **Markdown**.

   c (Optional) To enable **Implicit row detection**, select the toggle to on.

   d (Optional) To enable **Implicit column detection**, select the toggle to on.

   e (Optional) To **Detect borderless tables**, select the toggle to on.

   f (Optional) Enter a value for the **Table detection threshold**.

5 Click **Save** to save the configuration details.

The new configuration is added to the collection's **Summary** tab. To add more configurations to the collection, follow the steps above.

> **IMPORTANT** After you create a configuration, a good practice is to run a vectorization job on page 57. Doing so creates and saves the vector embeddings for your documents to the target vector database.

Configuring the LLMs

You can configure large language models (LLMs) to use within a collection or to use for evaluations. Once you configure the LLMs, then you can chat on page 113 with the configurations and run user evaluations on page 62.

Configure LLMs to Use within a Collection

1. Click to navigate to the Collections view.
2. In the Collections view, select the collection tile to configure LLMs. The collection opens to the **Summary** tab.

   TIP To find a specific collection, begin typing the collection name in the search field.

3. Select the **LLMs** tab.
4. Add or remove configurations.
   - To add a configuration, in the **Available large language Models** pane, click +› in each model that you want to add to the collection. The models are added to the **Configured large language models** pane.
   - To remove a configuration, in the **Configured large language models** pane, click ‹– in each model that you want to remove from the collection. The models are added to the **Available large language Models** pane.

   Note: for an available LLM means the connection succeeded. for an available LLM means the connection failed due to an error.

5. For each added model, select **Retrieval settings** from the settings that are available for each model.
6. Click to save the changes to the database.

   Note: To revert changes made to the collection, click

TIP You can reorder the list of configured LLMs. Click and hold in a configuration and move the model to the desired order. The order reflects the LLM priority that is seen by users in the Chat view. In that view, non-admin users can see and chat with only the first model in the ordered list. Admin users can see all configured models.

Configure LLMs for Evaluations

1. Click to navigate to the Collections view.
2. In the Collections view, select the collection tile that you want to configure LLMs for evaluations. The collection opens to the **Summary** tab.

> **TIP** To find a specific collection, begin typing the collection name in the search field.

3. Select the **LLMs** tab.
4. In the **User Eval** pane, select an **Eval LLM** from the menu to use with user evaluation tasks.
5. In the **Auto Eval** pane, select a **Data generation LLM**, **Critic LLM**, and an **Eval LLM** from the menus to use with auto evaluation tasks.

   Note: for an available LLM means the connection succeeded. for an available LLM means the connection failed due to an error.

6. Click to save the changes to the database.

   Note: To revert changes made to the collection, click

Managing Collections

The properties that you can specify to manage collections are described in "Details for a Collection" on page 40.

Run a Vectorization Job Manually

1. Click to navigate to the Collections view.
2. In the Collections view, select the collection tile that contains the configurations for which you want to run a vectorization.

> **TIP** To find a specific collection, begin typing the collection name in the search field.

3. On the collection's **Summary** tab, select a configuration.
4. Click ▶. The Submit Vectorization Job window opens.
5. (Optional) Review the default settings in the Submit Vectorization Job window, or select different values from the drop-down lists for the **CPU**, **Memory**, and **Execution provider**, and **OpenVINO device**.
6. Click **Submit**.

   The status updates in the Vectorization Job Status column as the job progresses.
7. (Optional) To review the log for the vectorization job, click ▤ in the configuration's Vectorization Job Status column.

Note: You can set a vectorization job to run automatically by using the automation settings in the Automation view. For more information about automation, see "About Automation" on page 107.

Change a Configuration's Status

1. Click ⧉ to navigate to the Collections view.
2. In the Collections view, select the collection tile that contains the configuration status that you want to change.

> **TIP** To find a specific collection, begin typing the collection name in the search field.

3. On the collection's **Summary** tab, select a configuration.
4. Click in the Configuration Status column for the configuration, and select **Under Evaluation**, **Challenger**, or **Champion**.

Modifying a Collection

Modify a Collection's Properties

1. Click to navigate to the Collections view.
2. Click in the collection tile that contains the properties to modify.
3. You can modify the collection's **Name** and **Description**. You cannot change the collection's sources.
4. Click **Save** to save your changes.

Modify a Collection's Configurations

You cannot modify a collection's configuration settings once you save it. You can make changes to a collection's configurations in the following ways:

- Copy the configuration and change specific settings on page 59.
- Add and delete a configuration in a collection on page 59.
- Add and delete LLMs for a collection on page 60.
- Change the LLMs for evaluations on page 60.

Copy a configuration and modify settings

1. Click to navigate to the Collections view.
2. On the Collections view, select the collection tile that contains the configuration that you want to copy.
3. On the **Sources** tab, select the configuration that you want to copy.
4. Click .
5. Modify the desired settings as described in "Create a Configuration" on page 54.

> **IMPORTANT** Be sure to run a vectorization job on page 57 on the new configuration.

Add and delete a configuration in a collection

> **IMPORTANT** You cannot delete a configuration if it is the only one in the collection and it is set as champion. You must create a new configuration and set the new configuration as the champion before you can delete the other

configuration. For steps to create a configuration, see "Create a Configuration" on page 54.

You cannot delete a configuration if it is being used for any evaluation runs or questions. You have to delete the evaluation information first. For steps to delete an evaluation, see "Delete a User Evaluation Test" on page 71. For steps to delete a question, see "Delete Automated Evaluation Questions" on page 65.

1 Click to navigate to the Collections view.

2 Select the collection for which you want to add or delete a configuration.

3 Add or remove a configuration on the **Summary** tab.

   - To add a configuration, see "Create a Configuration" on page 54.
   - To delete a configuration, select the configuration that you want to delete, click , and then click **OK** to confirm deletion.

> **IMPORTANT** If you add a configuration, be sure to run a vectorization job on page 57 on the new configuration.

Add and delete LLMs for collection

1 Click to navigate to the Collections view.

2 On the Collections view, select the collection tile to add or delete LLMs.

3 On the **LLMs** tab, add or remove models as described in "Configure LLMs to Use within a Collection" on page 56.

Change the LLMs for evaluations

1 Click to navigate to the Collections view.

2 On the Collections view, select the collection tile with the configurations that you want to change. The collection opens to the **Summary** tab.

3 On the **LLMs** tab, change the LLMs for user and auto evaluations as described in "Configure LLMs for Evaluations" on page 57.

View Existing Collections and Their Details

To see a list of the collections that have been created for SAS Retrieval Agent Manager, click to navigate to the Collections view. The existing collections are displayed as tiles that are organized in alphabetical order, from left to right. Each tile contains basic information, such as the collection name and its associated sources. To see the sources that are associated with a collection, click **Sources** in a collection's tile.

To see configuration details and evaluation test results for a specific collection, click the name of a collection in the Collections view. The information that is available on each tab of the collection is described in "About Collections" on page 39.

> **TIP** To find a specific collection, begin typing the collection name in the search field in the Collections view.

Delete a Collection

> **IMPORTANT** You cannot delete a collection that has any configurations that are being used by an agent. Before you delete the collection, you must modify the agent by assigning it to a different collection. For steps to modify an agent, see "Modify an Agent" on page 104.
>
> You cannot delete a collection if any of its configurations are being used for any evaluation runs or questions. You have to delete the evaluation information first.
>
> You cannot delete a configuration if it is the only one in the collection and it is set as champion. You must create a new configuration and set the new configuration as the champion before you can delete the other configuration. For steps to create a configuration, see "Create a Configuration" on page 54.

1. Click to navigate to the Collections view.
2. In the Collections view, click in the collection tile that you want to delete.

   > **TIP** To find a specific collection, begin entering the collection name in the search field.

3. Click .
4. Click **OK** to verify the action.

   The collection is removed from the Collections view.

Managing User Evaluations

The properties that you can specify to manage user evaluations are described in "User Eval Tab" on page 46.

Run a New User Evaluation

Note: Only authorized users can run a new user evaluation.

1. Click to navigate to the Collections view.
2. In the Collections view, click the collection tile that contains the user evaluation to run.

   TIP To find a specific collection, begin typing the collection name in the search field.

3. On the **User Eval** tab, click . The **New Evaluation** window opens.
4. On the **Configurations** tab, select one or more available configurations and click +›. The configurations are added to the right pane.

   To configurations from the right pane, select one or more files, click ‹+.

5. On the **Tests** tab, select the test prompt to use. The prompts that the model looks for are listed under **Test prompts**.
6. Click **OK**. The job is added to the evaluations table and starts.

   You can track the job's status in the Status column. next to the status means that the job ran successfully. next to the status means that the job failed due to an error. In the Status column, you can view the log for that job.

View Details about User Evaluation Jobs

To see details about user evaluation jobs for a collection, click to navigate to the Collections view. Click the collection tile that contains the user evaluation data that you want to view. Select the **User Eval** tab. The user evaluations that have run are listed in the table at the top of the tab. When you select a job in the table, the details for the run are displayed below the table.

You can select each **User eval prompt** that is listed for the job to see its details. Details include the full text of the prompt, the response, and the assertions for the response. You can see the details of each metric in the **Assertions** section and whether the assertion passed.

The information that is displayed on this tab is described in "User Eval Tab" on page 46.

Delete User Evaluation Job Results

Note: Only authorized users can delete user evaluation job results.

1 Click to navigate to the Collections view.

2 in the Collections view, click the collection tile that contains the user evaluation results to delete.

> **TIP** To find a specific collection, begin typing the collection name in the search field.

3 On the **User Eval** tab, select the user evaluation job that you want to delete.

4 Click , and click **OK** to confirm the action.

The job is removed from the user evaluation jobs table.

Managing Auto Evaluations

The properties that you can specify to manage auto evaluations are described in "Auto Eval Tab" on page 42.

Generate Auto-Evaluation Data

1 Click to navigate to the Collections view.

2 In the Collections view, click the collection tile that contains the auto-evaluation job to run.

> **TIP** To find a specific collection, begin entering the collection name in the search field.

3 On the **Auto Eval** tab, click above the Auto Eval Data table. The Generate Evaluation Data window opens.

4 (Optional) You can accept the current value for **Source coverage** or enter a different value.

The value for **Num test cases to be generated** changes automatically depending on the value for **Source coverage**.

5 (Optional) You can accept the current value for **Reference configuration** or select a different value.

6 Click **OK** to submit the data generation job.

The **Data generation jobs** progress indicator spins as the job runs. You can expand the drop-down menu for **Data generation jobs** to see the job's progress and a list of previous jobs. next to a data generation job means that the job ran successfully. next to a data generation job means that the job failed due to an error. To view the log for a job, select a job from the **Data generation jobs** drop-down menu and select **View Logs**.

When the job finishes, the generated questions and ground truth answers are added to the Auto Eval Data table. You can delete generated questions that are not useful. For more information, see "Delete Automated Evaluation Questions" on page 65.

A good practice is to next ask the pipeline the same questions to see what answers are generated. For more information about how to run an auto-evaluation job, see "Run an Automated Evaluation Job" on page 64.

Run an Automated Evaluation Job

Note: Only authorized users can run an automated evaluation job.

When you run an automated evaluation job, questions are passed to the RAG pipeline. The results are compared to the ground truth answers by using a *critic large language model* (LLM). For more information about the metrics used, see "Auto Eval Tab" on page 42.

Here is how to run an automated evaluation job:

1 Click to navigate to the Collections view.

2 In the Collections view, click the collection tile that contains the automated evaluation job to run.

> **TIP** To find a specific collection, begin typing the collection name in the search field.

3 On the **Auto Eval** tab, click above the Auto Eval Results table.

4 (Optional) In the Run Evaluation window, select a different **Critic Embedding Model** from the list of available models.

5 (Optional) In the **Configurations** field, add or remove configurations from the run.

- To add a configuration, select one or more configurations in the left pan and click +›. The models are added to the right pane.
- To remove a configuration, select one or more configurations in the right pan and click ‹+. The models are added to the left pane.

6 Click **OK**.

The jobs are added to the Auto Eval Results table.

View Automated Evaluation Data

To see the data for automated evaluations that exist for a collection, click to navigate to the Collections view. Click the collection tile that contains the automated evaluation data that you want to view. Select the **Auto Eval** tab. The Auto Eval Data table lists the generated questions and information about the evaluation data. To see the status of the last run, click **Data generation jobs**.

The Auto Eval Results table lists the automated evaluation jobs that have run and metrics about each run. The **Filter stale results** toggle enables you to remove automated evaluation results for jobs that are not in sync with the current automated evaluation data (for example, a new automated evaluation question was added or deleted). In addition, to see detailed metrics and outputs for each question in a job, you can click **View** in the Details column for a specific job to open its Evaluation Results window.

The information that is displayed in the tables on this tab is described in “Auto Eval Tab” on page 42.

Delete Automated Evaluation Questions

Note: Only authorized users can delete automated evaluation questions.

An LLM might generate a question that is not useful to your evaluations. In this case, you can delete the question from your evaluation.

1 Click to navigate to the Collections view.

2 In the Collections view, click the collection tile that contains the automated evaluation question to delete.

> **TIP** To find a specific collection, begin typing the collection name in the search field.

3 On the **Auto Eval** tab, select the question to delete from the Auto Eval Data table.

4 Click .

5 Click **OK** to verify the action.

The question is removed from the Auto Eval Data table.

7

Working with User Evaluation Tests

About User Evaluation Tests

Note: This view is enabled only for authorized users.

Note: For an overview of evaluations, see "Overview of Evaluations in SAS Retrieval Agent Manager" on page 7.

Evaluation tests are reusable scenarios that you define to use for manual evaluations. Evaluation tests are not associated with a specific collection or agent. They can be used on any agent or collection in the system.

In the User Eval Tests view, you can see a list of the existing user evaluation tests, and you can create, modify, and delete tests and their associated test prompts. To create a user evaluation test, you define the properties for the test scenario, set a minimum passing threshold, and create one or more test prompts and expected assertions.

The following table describes the details about existing user evaluation tests.

Table 7.1 Details about a User Evaluation Tests

Field	Description
Name	The name of the test. Example: `Financial questions`
Passing Threshold	The number of assertions as a percentage that have to pass in order for a response to be considered an appropriate answer to a prompt. Example: `90`
Updated At	The datetime at which the test was last updated. Example: `May 30, 2035, 3:00:00 AM EST`

The following table describes the details about the test prompts for a user evaluation test.

Table 7.2 Details about the Test Prompts for a User Evaluation Test

Field	Description
Test prompts	The list of prompts written for a user evaluation test. Example: `What products does Apple make?`
Prompt	The text of the test prompt that is selected. Example: `What products does Apple make?`
Passing threshold	The number of assertions as a percentage that have to pass in order for a response to be considered an appropriate answer to a prompt. Example: `100`
Assertions	The specifications that were made for the prompt, which include a Metric (for example, contains) and the Value (for example, `I don't know`).

Create a User Evaluation Test

The properties that you can specify when you create a user evaluation test are described in “About User Evaluation Tests” on page 67.

To create a user evaluation test, you define the properties for the test and then create one or more test prompts to run against your models.

Define the properties

1. Click to navigate to the User Eval Tests view.
2. Click .
3. In the New User Eval Test window, enter a **Name** for the test. Test names should be unique.
4. (Optional) Enter a **Description** of the test.
5. Select or enter a value for the **Passing Threshold**.
6. Click **OK** to save the test. The test is added to the list of user evaluation tests in alphabetical order .

Create the test prompts to run against models

1. To create a test prompt for the new user evaluation test, click **Create one**.
2. In the **Prompt** field, enter your test question.
3. Enter the **Passing threshold (%)**.
4. Under **Assertions**, select the **Metric**, and enter a **Value**.
5. (Optional) To add another assertion, click **New assertion**, select the **Metric**, and enter a **Value**.
6. Click to save the test prompt.

> **TIP** To revert the last change made to the test prompt, click .

7. (Optional) To add another test prompt to the user evaluation test, click above the **Prompt** field and repeat the steps in this section, starting with Step 2 on page 69.

A good practice is to apply safeguard prompts. For example, to ensure that the large language model (LLM) does not provide unrealistic answers, write a prompt for the LLM to respond "I do not know" to real-time queries.

Modify a User Evaluation Test

Modify the Properties for an Evaluation Test

1. Click to navigate to the User Eval Tests view.

2 Select the name of the user evaluation test that you want to modify.

3 Click above the list of user evaluation tests.

4 You can modify the **Name**, **Description**, and **Passing theshold (%)** as described in “Define the properties” on page 69.

5 Click **OK** to save your changes.

Modify the Prompts for an Evaluation Test

1 Click to navigate to the User Eval Tests view.

2 Select the name of the user evaluation test with the prompts that you want to modify.

3 In the **Test prompts** list, select the prompt that you want to modify.

4 In the **Prompt** and **Passing threshold (%)** fields, you can enter new values. In the **Assertions** section, you can select a different **Metric** and change the **Value**.

5 You can add and remove a test prompt.

- To add a test prompt, see “Create the test prompts to run against models” on page 69
- To remove a test prompt, select the prompt to remove from the **Test prompts** list and click .

6 You can add and remove an assertion.

- To add an assertion, click **New assertion**, select the **Metric**, and enter a **Value**.
- To remove an assertion, click in the row that contains the metric and value to remove.

View Existing User Evaluation Tests and Their Details

To see a list of the user evaluation tests that have been created for SAS Retrieval Agent Manager, click to navigate to the User Eval Tests view. The existing user evaluation tests are listed in this view. By default, items are sorted alphabetically by the Name column. You can sort the list based on values in any column or enter a string in the search field to find a specific item.

To see the prompts that are associated with a specific user evaluation test, select the name of a test from the User Eval Tests table. The prompts that are associated with that test are listed under **Test prompts**. You can select a test prompt to see its details in the right pane.

> **TIP** To find a specific user evaluation test, begin typing the test name in the search field in the User Eval Tests view.

The properties that are shown in the User Eval Tests view are described in "About User Evaluation Tests" on page 67.

Delete a User Evaluation Test

1 Click to navigate to the User Eval Tests view.

2 Select the evaluation test that you want to delete.

> **TIP** To find a specific test, begin typing the name in the search field.

3 Click .

4 Enter the confirmation code that is displayed in the Delete test window.

5 Click **OK** to verify the action.

The test is removed from the list of user evaluation tests.

8

Working with Code Templates

About Code Templates

Note: This view is enabled only for authorized users.

The Code Templates view in SAS Retrieval Agent Manager enables you to create and manage templates that simplify integration and customization tasks. A template is a predefined configuration blueprint that defines the structure and settings for its respective component. Templates help to ensure consistency and to reduce setup time. Here are the available template types:

- MCP tool server template

  This template defines connection parameters for MCP tools, default properties (such as tool grouping, resource paths, and security settings), and integration hooks for SAS Viya or other SAS environments.

- agent template

  This template provides a starting point for creating retrieval agents with standardized configurations. It helps streamline agent setup and ensures alignment with organizational best practices.

- orchestrator agent template

  This orchestrator agent template offers a standardized foundation for building agents. It helps streamline configuration and maintain consistency with organizational best practices.

- custom source template

  This template enables you to define custom data sources for retrieval workflows. It is useful when integrating nonstandard or proprietary data sources into SAS Retrieval Agent Manager.

To begin integrating MCP tools with your agents, see “Create an MCP Tool Server Template” on page 74. For instructions on creating agent templates or custom source templates, see “Create an Agent Template” on page 76 or “Create a Custom Source Template” on page 80.

Create an MCP Tool Server Template

The first step to integrating MCP tools with your agents is to create an MCP tool server template.

1 Click </> to navigate to the Code Templates view.

2 Create an MCP tool server template from one of the following options:

- From Python code: click **Code MCP Server**
- From a container image: click **Container MCP Server**

Code MCP Server

1 On the Create new code tool template window, enter a name for the server template.

Note: A best practice is to name the server template based on its function. Do not include sensitive data in the name. In addition, the name should not contain version or environment information.

2 Enter a description.

Note: Although a description is not required, a best practice is to briefly describe what types of tools the code template contains.

3 Click the **Code** tab.

4 On the **run.py** tab, enter the Python script for the MCP tools that you want to include in the server template.

5 Click the **requirements.txt** tab.

6 On the **requirements.txt** tab, add your tools' Python dependencies.

7 If your Python script includes variables, click the **Environment Variables** tab.

8 Click .

9 Click the new row that was added to the table.

10 For **Name**, enter the name of the variable, then press Enter.

11 For **Description**, describe what the variable represents, then press Enter.

12 For **Default Value**, enter the default value for the variable, then press Enter.

13 If the variable consists of sensitive information, select the check box in the Secret column.

14 Click **OK**.

Container MCP Server

1 On the Create new container tool template window, enter a name for the server template.

Note: A best practice is to name the server template based on its function. Do not include sensitive data in the name. In addition, the name should not contain version or environment information.

2 Enter a description.

Note: Although a description is not required, a best practice is to briefly describe what type of tools the code template contains.

3 For **Container image**, enter the full name and tag of the container image that contains the MCP server implementation for the tools that you want to include in the server template.

4 If user authentication is required to access the container image, enter your user name and password associated with your container registry account.

5 If the container image requires command-line arguments to be passed to the container when it runs, enter the arguments.

6 For **Transport**, select the communication protocol or mechanism used to send requests and receive responses.

7 For **Port**, enter the network port that the containerized tool exposes for communication.

8 For **Base Path**, enter the file path for the root directory inside the container.

9 For **Requested CPU**, specify how much CPU resource should be allocated to the container when it runs.

10 For **Requested Memory**, specify the amount of RAM that the container should reserve when it runs.

11. If the Python script for the containerized tool includes variables, click the **Environment Variables** tab.
12. Click .
13. Click the new row that was added to the table.
14. For **Name**, enter the name of the variable, and then press Enter.
15. For **Description**, describe what the variable represents. and then press Enter.
16. For **Default Value**, enter the default value for the variable. and then press Enter.
17. If the variable consists of sensitive information, select the check box in the Secret column.
18. Click the **Configuration File** tab.
19. On the **Configuration File** tab, specify the absolute path inside the container where the configuration file is located.
20. For **Content**, copy and paste the content of the configuration file to be mounted in the container.
21. Click **OK**.

For the next steps in setting up your MCP tools integration, see “Publish and Unpublish a Template” on page 81.

Create an Agent Template

1. Click </> to navigate to the Code Templates view then click **Code Agent**.
2. On the **Details** tab, enter a name and description for the agent template.

   Note: A best practice is to enter a unique, descriptive name. Do not include sensitive data in the name. In addition, the name should not contain version or environment information. Although a description is not required, a best practice for the description is to briefly summarize information about the template such as when and why someone would use it and include any limitations or assumptions.

3. On the **Code** tab, click **Browse** to select the ZIP package that contains the run.py and requirements.txt files for the agent. If necessary, you can use the code box to edit the code so that it is generalized for template reuse. For example, you should remove hardcoded references such as file paths, credentials, ports, API endpoints, and direct URLs and replace them with environment variables. You should also use aliases to refer to LLMs, tools, and collections.

If you do not already have a ZIP package that contains the run.py and requirements.txt files for the agent, then you can click **+** to create the files with the code editor. When you create the run.py file using this method, some of the code is automatically populated to help get you started.

> **IMPORTANT** Do not include confidential values such as credentials in your code. Instead, add them to the template as environment variables.

CAUTION

Do not run unsafe code. Running unsafe code can expose your environment to data loss, data corruption, security breaches, or system instability. It is your responsibility to ensure that all code that you incorporate is secure, tested, and compliant with your organization's policies and applicable law.

Example 1: The following code sample in the run.py file enables you to specify that responses to queries are returned in Spanish.

```
import time
from typing import Dict

def exec(
    text: str,
    client,
    options
) -> str:
    """
    Demo exec function showcasing use of the client library.
    """
    print(f"Received request: {text}")
    print(f"Plugin options: {options}")

    # Returns a `QueryRequest` object which can be used later to fetch
    # the query result.
    # Asynchronous query example.
    query_request_1 = client.post_query_async(
        prompt=text,
        collection_name="default",
        llm_name="default",
    )

    response = query_request_1.get_result(timeout=60)  # Blocks until
query is finished or the request times out
    answer = response.response.answer
    print(f"Query finished: {answer}")

    # Now pass this response to the LLM to convert it to Spanish.
    # Synchronous query example.
    response = client.post_query(
        prompt=f"Convert the following text to Spanish: {answer}",
        llm_name="default",
        direct_llm_query=True,
    )
    answer = response.response.answer
```

```
    return answer
```

Example 2: The following code sample in the config.json file shows how you can define multiple LLMs that your agent can use.

```
{
    "options": [
        {
            "name": "Retry limit",
            "type": "number",
            "description": "Maximum tries for the LLM to attempt valid
XML generation",
            "default": 5
        },
        {
            "name": "LLM",
            "type": "string",
            "description": "LLM to use",
            "default": "gpt4o"
        }
    ],
    "extraCollections": [
    ],
    "extraLLMs": [
        {
            "alias": "gpt4o",
            "description": "GPT4o model alias"
        },
        {
            "alias": "local",
            "description": "Local LLM alias"
        }
    ]
}
```

Example 3: The following code sample in the config.json file shows how you can specify more than one parameter for an agent.

```
{
    "options": [
        {
            "name": "Tickers",
            "type": "string",
            "description": "Comma-separated string of tickers to
track",
            "default": "AAPL",
            "value": "AAPL,MSFT,AMZN"
        },
        {
            "name": "EmailList",
            "type": "string",
            "description": "Comma-separated string of emails",
            "default": "user1@domain-name.com",
            "value": ""
        }
    ]
}
```

Example 4: The following is a sample requirements.txt file with the names of Python packages to install.

```
# Agent-specific dependencies.
matplotlib
yfinance
```

4 On the **Environment Variables** tab, for each environment variable that you implemented, click then add a name, description, and default value.

5 On the **Aliases** tab, for each alias that you used, click then enter the alias for the LLM, tool, or collection and provide a brief description of the component.

6 Click **Save**.

For next steps, see "Publish and Unpublish a Template" on page 81.

Create an Orchestrator Agent Template

1 Click </> to navigate to the Code Templates view and then click **Code Orchestrator Agent**.

2 On the **Code** tab, click **Browse** to select the ZIP package that contains the run.py and requirements.txt files for the agent. If necessary, you can use the code box to edit the code so that it is generalized for template reuse. For example, you should remove hardcoded references such as file paths, credentials, ports, API endpoints, and direct URLs and replace them with environment variables. You should also use aliases to refer to LLMs, tools, and collections.

If you do not already have a ZIP package that contains the run.py and requirements.txt files for the agent, then you can click + to create the files with the code editor. When you create the run.py file using this method, some of the code is automatically populated to help get you started.

> **IMPORTANT** Do not include confidential values such as credentials in your code. Instead, add them to the template as environment variables.

CAUTION

Do not run unsafe code. Running unsafe code can expose your environment to data loss, data corruption, security breaches, or system instability. It is your responsibility to ensure that all code that you incorporate is secure, tested, and compliant with your organization's policies and applicable law.

Example 1: The following code sample in the run.py file creates a basic orchestrator agent.

```
from langchain_core.messages import HumanMessage, SystemMessage
from sasram.agent import OrchestratorClient
```

```
#Creates an orchestrator agent that has tools to call the agents that
it has access to
async def exec(text: str, ram_client: OrchestratorClient) -> str:
    agent = await ram_client.get_langgraph_agent()

    #Loads the session history and combines that with the user's query
and a system message telling the ocrchestrator agent to use its agents
to inform its response
    session_history = ram_client.get_session_history_as_langchain()
    messages = (
        [SystemMessage("Use your sub-agents to inform your
responses.")]
        + session_history
        + [HumanMessage(content=text)]
    )
    #Invokes the orchestrator agent
    response = await agent.ainvoke({"messages": messages})
    #Returns the orchestrator agent's response
    return response["messages"][-1].content
```

3. On the **Environment Variables** tab, for each environment variable that you implemented, click and then add a name, description, and default value.

4. On the **Aliases** tab, for each alias that you used, click and then enter the alias for the LLM, tool, or collection and provide a brief description of the component.

5. Click **Save**.

For next steps, see "Publish and Unpublish a Template" on page 81.

Create a Custom Source Template

1. Click </> to navigate to the Code Templates view then click **Custom Source**.

2. On the **Details** tab, enter a name and description for the custom source template and select whether to enable LLM Aliases.

   Note: A best practice is to enter a unique, descriptive name. Do not include sensitive data in the name. In addition, the name should not contain version or environment information. Although a description is not required, a best practice for the description is to briefly summarize information about the template such as when and why someone would use it and any limitations or assumptions.

   When LLM aliases are enabled, it lets your custom source template reference LLMs in its code using stable, human-readable names instead of IDs that are subject to change. This avoids code changes when LLM configurations change. It also enables the use of multiple LLMs to support different use-cases within a custom source.

3 On the **Code** tab, click **Browse** to select the Python file or zip package that contains the code for the custom source. If necessary, you can use the code box to edit the code so that it is generalized for template reuse. For example, you should remove hardcoded references such as file paths, credentials, API endpoints, and direct URLs and replace them with environment variables.

> **IMPORTANT** The Python code for a custom source template must include the `exec()` function. In addition, the code should not include confidential values such as credentials in your code. Instead, add them to the template as environment variables.
>
> For information about Python, see Python documentation.

4 On the **Environment Variables** tab, for each environment variable that you implemented, click then add a name, description, and default value.

5 If you used aliases to reference LLMs in the code for the custom source template, go to the **Aliases** tab and, for each alias that you used, click then enter the LLM alias and provide a brief description of the LLM.

6 Click **Save**.

For next steps, see “Publish and Unpublish a Template” on page 81.

Publish and Unpublish a Template

Publishing a template makes that template available in the list box when adding or modifying an agent, custom source, or tool server. Unpublish a template to remove that template from the list box.

> **IMPORTANT** You can unpublish and republish the latest version of a template as needed until a newer version is created. After a newer version is created, if you unpublish an older version of the template, you cannot republish the older version.

1 Click </> to navigate to the Code Templates view.

2 Select the template that you want to publish or unpublish.

3 To publish the template, click .

If you are setting up your MCP tool integration, see “About MCP Tools” on page 85 for next steps.

4 To unpublish the template, click .

Version a Template

You cannot modify an existing version of a template. However, you can create a new version of the template. To create a new version of a template, complete the following steps.

1. Click </> to navigate to the Code Templates view.
2. Select the template that you want to version.
3. Modify properties on the **Settings**, **Code**, and **Environment Variables** tabs as described in the associated topic.
   - “Create an MCP Tool Server Template” on page 74
   - “Create an Agent Template” on page 76
   - “Create a Custom Source Template” on page 80
4. Click .
5. On the Save window, if you want to make the template available for selection in the associated view immediately after saving, select **Publish on save**.
6. Click **OK** to save your changes as a new version of the template.

   Note: If you did not select **Publish on save**, when you are ready to make the template available, click .

Delete a Template

Deleting a template permanently removes the template and its version history from the application. To delete a template, complete the following steps.

> **IMPORTANT** You cannot delete an MCP tool server template that is currently in use by a tool server. You must stop the tool server and then modify it by selecting a different tool server template.
>
> For steps to stop the tool server, see “Start and Stop an MCP Tool Server” on page 90. For steps to modify the tool server, see “Modify an MCP Tool Server” on page 90.

1. Click </> to navigate to the Code Templates view.
2. For the template that you want to delete, click ⋮ and then select **Delete**.
3. Click **OK**.

View Existing Templates and Their Details

1. Click </> to navigate to the Code Templates view.

   Existing templates are displayed in the bottom pane.
2. To view details about a template, click the template name.
3. In the top right corner, select the version of the template for which you want to view details.
4. View template properties on the **Settings**, **Code**, and **Environment Variables** tabs.

9

Working with MCP Tools

About MCP Tools

Note: This view is enabled only for authorized users.

CAUTION
Connect only to tool servers from secure, trusted sources. Connecting to tool servers from unsecure or untrusted sources can expose your environment to data exfiltration, data corruption, security breaches, or system instability. Customers are responsible for ensuring that all tool server connections are secure, tested, vetted under applicable internal security policies, and compliant with applicable law. SAS makes no representations regarding the security, reliability, or content of third-party tool servers.

After you set up a tool server template on the Code Templates view, the next step to integrating MCP tools with your agents is to add an MCP tool server. The MCP Tools view enables you to create and manage Model Context Protocol (MCP) tool servers in SAS Retrieval Agent Manager. MCP tool servers act as hosts for MCP tools and resources. They provide a secure runtime environment and API endpoints for your agents to interact with external MCP tools, data sources, and services to perform specific tasks. Here are some examples of tasks that agents can perform using MCP tools include:

- fetching real-time data such as weather, financial metrics, and IoT sensor readings
- creating and editing files
- running predictive models
- performing streaming analytics
- running scripts or code snippets

For steps to add an MCP tool server, see "Adding an MCP Tool Server" on page 86.

Adding an MCP Tool Server

CAUTION

Connect only to tool servers from secure, trusted sources. Connecting to tool servers from unsecure or untrusted sources can expose your environment to data exfiltration, data corruption, security breaches, or system instability. Customers are responsible for ensuring that all tool server connections are secure, tested, vetted under applicable internal security policies, and compliant with applicable law. SAS makes no representations regarding the security, reliability, or content of third-party tool servers.

After you have added an MCP tool server as described in the following sections, see "Start and Stop an MCP Tool Server" on page 90 for the next steps in your MCP tool integration.

Add an MCP Tool Server from a Catalog Template

1. Click to navigate to the MCP Tools view.
2. Click .
3. On the New Tool Server window, enter a name for the tool server.

   Note: A best practice is to name the tool server based on its function. Do not include sensitive data in the name. In addition, the name should not contain version or environment information.

4. Enter a description.

   Note: Although a description is not required, a best practice is to briefly define what the tool server does.

5. Select **Catalog**.
6. Select one of the following template types.

- **DB Connector**
- **OpenAPI MCP Server**
- **Remote MCP Server**

7 Click **Select Template**.

8 If a template is already configured, you can select the template name and version.

If a template is not already configured, complete the following steps for the option that you selected.

DB Connector

To configure a **DB Connector** template:

1 For **Vendor**, select the remote database that you want to connect to.

2 For **Version**, select the version of the template that you want to use.

3 Click **Next**.

4 On the **Environment Variables** tab, define the variables.

5 Click **Finish**.

OpenAPI MCP Server

To configure an **OpenAPI MCP Server** template:

1 Click **Next**.

2 On the **Connection** tab, enter the URLs for the MCP tool server and the OpenAPI specifications.

3 Click **Next**.

4 On the **Authentication** tab, select an authentication type.

Authentication Type	Configuration
None	No further configuration is needed.
API key	Select the **Use RAM credentials** option to use your SAS Retrieval Agent Manager token or enter your API key for the MCP server. **TIP** Select **Use RAM credentials** if your organization is set up to use single sign-on.
OAuth client credentials	Provide these credentials: ■ Your OAuth client ID ■ Your OAuth client secret ■ Your OAuth token URL

Authentication Type	Configuration
	■ Your OAuth scope

5 Click **Finish**.

Remote MCP Server

To configure a **Remote MCP Server** template:

1 Click **Next**.

2 On the **Connection** tab, enter the transport type and the URL for the MCP tool server.

3 Click **Next**.

4 On the **Authentication** tab, select an authentication type and complete the associated configuration.

Authentication Type	Configuration
None	No further configuration is needed.
API key	Select the **Use RAM credentials** option to use your SAS Retrieval Agent Manager token or enter your API key for the MCP server. **TIP** Select **Use RAM credentials** if your organization is set up to use single sign-on.
OAuth client credentials	Provide these credentials: ■ Your OAuth client ID ■ Your OAuth client secret ■ Your OAuth token URL ■ Your OAuth scope

9 Click **Finish**.

Add an MCP Tool Server from a Custom Template

1 Click to navigate to the MCP Tools view.

2 Click .

3 On the New Tool Server window, enter a name for the tool server.

Note: A best practice is to name the tool server based on its function. Do not include sensitive data in the name. In addition, the name should not contain version or environment information.

4 Enter a description.

Note: Although a description is not required, a best practice is to briefly define what the tool server does.

5 Select **Custom**.

6 Select the template name and template version that the tool server should use.

7 Click **Select Template**.

8 If applicable, click **Next**.

Depending on the template set up in the Code Templates view, you might need configure either or both of the following tabs.

a On the **Authentication** tab, select an authentication type and complete the associated configuration.

Authentication Type	Configuration
None	No further configuration is needed.
API key	Select the **Use RAM credentials** option to use your SAS Retrieval Agent Manager token or enter your API key for the MCP server. **TIP** Select **Use RAM credentials** if your organization is set up to use single sign-on.
OAuth client credentials	Provide these credentials: ▪ Your OAuth client ID ▪ Your OAuth client secret ▪ Your OAuth token URL ▪ Your OAuth scope

b On the **Environment Variables** tab, define the variables.

9 Click **Finish**.

Start and Stop an MCP Tool Server

When you start an MCP tool server, the tools that it contains become available on the **Tools** tab on the Add Agent and Edit Agent windows. When you stop an MCP tool server, the tools become unavailable. To start or stop an MCP tool server, complete the following steps.

1 Click to navigate to the MCP Tools view.

2 Select an MCP tool server from the list.

3 Check the state of the server in the Server Status column.

4 To start the server, click .

If you are setting up your MCP tool integration, see "Add an Agent" on page 96 for next steps.

5 To stop the server, click .

Modify an MCP Tool Server

> **IMPORTANT** You cannot modify an MCP tool server that is running. For steps to stop the tool server, see "Start and Stop an MCP Tool Server" on page 90.

1 Click to navigate to the MCP Tools view.

2 Select an MCP tool server from the list.

3 Click .

4 On the Edit window, you can modify properties on the **Settings**, **Environment Variables**, and **Files** tabs as described in "Adding an MCP Tool Server" on page 86.

5 Click **OK**.

Delete an MCP Tool Server

When you delete an MCP tool server, it is permanently removed from the application and all associated tools are deleted. To delete a tool server, complete the following steps.

> **IMPORTANT** You cannot delete an MCP tool server that is running or that an agent is using. If an agent uses a tool from the MCP tool server that you want to delete, first remove the tool from the agent, then stop the tool server. For steps to edit an agent, see "Modify an Agent" on page 104. For steps to stop the tool server, see "Start and Stop an MCP Tool Server" on page 90.

1. Click to navigate to the MCP Tools view.
2. In the **MCP Tool Servers** table, select the tool server that you want to delete.
3. Click .
4. Click **OK**.

View Existing MCP Tool Servers and Their Details

1. Click to navigate to the MCP Tools view.

   On the MCP Tools view, the **MCP Tool Servers** table displays the following details about each tool server:

   - the tool server name
   - the code template associated with the tool server
   - the server status
   - the ping status

   For tool servers that are currently running, is displayed next to the server status.

2. Click to display server activity on the Tool Server Logs window.

3. For more details about a tool server, select the tool server's name in the **MCP Tool Servers** table.

   If the tool server is currently running, the bottom pane lists the tools contained in the selected tool server, alongside descriptions of what each tool does.

4. In the bottom pane, click > next to a tool name to display parameter details for that tool.

   Parameter details include:

   - the parameter name
   - the field type for the parameter
   - whether the parameter is required
   - the default value for the parameter

10

Working with Agents

About Agents

Note: This view is enabled only for authorized users.

There are two types of agents in SAS Retrieval Agent Manager: agents and orchestrator agents.

Orchestrator agents are task delegators that dissect queries into separate tasks and assign those tasks to agents. When you chat with an orchestrator agent, the agent analyzes your prompt, determines which tasks are required, and routes each task to the most appropriate agent based on the information exposed in the agent's A2A agent card.

Agents are task orchestrators that enable you to extend the core retrieval-augmented generation (RAG) capabilities by using custom Python workflows. Unlike basic chat interactions, agents enable more complex automations, such as the following:

- Use custom parameters and logic to tailor responses.
- Execute Python scripts to handle specialized or multistep workflows.
- Retrieve information from databases.
- Submit API requests.

Each agent follows a defined workflow to execute a sequence of operations that generate intelligent responses or automate decision-making. This workflow is specified in custom Python code that is added when you create an agent template. In addition, because agents are fully hosted within a Kubernetes cluster, they do not require external deployment and can be queried from within the system.

Agents View

The Agents view lists the existing agents that execute workflows within the RAG pipeline. Agents are organized by name in alphabetical order. This view enables you to create, edit, start, stop, and delete agents. You can also view an agent's log to get visibility into run-time behaviors or issues.

The following table describes the information that is displayed in the Agents view about each existing agent.

Table 10.1 *Information about Existing Agents in the Agents View*

Property	Description
Name	The name of the agent. Example: `Financial agent`
Description	The description of the agent.
Experiments	The number of experiments created for the agent.
Status	The running status of the agent. The status can be **Running**, **Stopped**, **No Champion**, or **Error**. Example: `Running`
Last Modified	The date and time at which the agent was last modified.

Create New Agent Window

When you create an agent, you can specify the agent name, provide a description of the agent, and enter a chat initializer command.

Property	Description
Name	The name of the agent. Example: **`Wall Street Journal agent`**
Description (Optional)	A string that describes the agent.
Chat initializer command (Optional)	A command that is sent to the agent to execute specific instructions when a chat session is initialized. Example: **`#summarizekeydevelopments`**

Create New Orchestrator Agent Window

When you create an orchestrator agent, you can specify the agent name, provide a description of the agent, and enter a chat initializer command.

Property	Description
Name	The name of the agent. Example: **`Financial news agent`**
Description (Optional)	A string that describes the agent.
Chat initializer command (Optional)	A command that is sent to the agent to execute specific instructions when a chat session is initialized. Example: **`#sendnewsletter`**

Edit Agent Window

The **Edit Agent** window contains the **Summary** tab and the **User Evaluation** tab. On the **Summary** tab, you can edit the agent's name, description, and chat initializer command. You can also view, create, edit, and delete agent experiments. On the **User Evaluation** tab, you can run user evaluation tests against agent experiments.

Add an Agent

The properties that you can specify when you create an agent or an orchestrator agent are described in "About Agents" on page 93.

1 Click to navigate to the Agents view.

2 Click **Create an Agent** or **Create an Agent Orchestrator**.

3 Enter a name for the agent.

Note: A best practice is to enter a unique, descriptive name. Do not include sensitive data in the name. In addition, the name should not contain version or environment information.

4 Enter a description of the agent.

Note: A best practice for the description is to briefly summarize information about the agent such as when and why someone would use it and any limitations or assumptions.

5 (Optional) Enter the system prompt that you want to use for all of the agent's chat sessions.

Note: If you enter a system prompt for an agent, the prompt overrides any system prompt configured in the Retrieval Settings view.

6 Enter a chat initializer command if you want the agent to execute specific instructions when a chat session is initialized.

7 Click **Create**.

The new agent is added to the list of agents in the Agents view.

By default, when you add a new agent, its state in the Status column is **No Champion**. For next steps, see "Creating and Managing Agent Experiments" on page 97.

Configure an A2A Agent Card

When you query an orchestrator agent, the orchestrator agent reads the information in their assigned agents' A2A agent cards to determine which agent has the appropriate tools to complete the associated task.

1. Click to navigate to the Agents view.
2. Select an agent and then click the **A2A Agent Card** tab.
3. If a description does not already exist, enter a description of the agent. The description should indicate the intended purpose of the agent. For example, for an agent called New York Times News Agent, you might enter **`an agent that retrieves and summarizes the latest headlines from the New York Times magazine.`**
4. If this is the first skill for the agent, click **Add Skill**

   If other skills already exist in the **Skills** section, click **+**.
5. Enter an informative name and description that explain what actions the agent can perform through its assigned tools.
6. In the **Query Examples** section, click **+** and then enter examples of prompts for which the agent is specialized to answer. The orchestrator agent uses these prompt examples to determine when and how to use the agent.
7. Click **Save Card**.

Creating and Managing Agent Experiments

Create an Agent Experiment for an Agent

Before you can interact with a new agent in the Chat view, you must create one or more agent experiments. Agent experiments let you create and evaluate different agent configurations so that you can determine which one performs best before selecting it as the champion configuration. After a champion is selected, you can start the agent. Starting the agent makes it available in the Chat view.

> **IMPORTANT** Before you can create an agent experiment, you must have at least one collection with a configuration selected as champion on the **Summary** tab in the Collections view.

1 Click to navigate to the **Agents** view.

2 Double-click the name of the agent for which you want to create an experiment.

3 On the **Summary** tab, if this is the first agent experiment for the agent, click **Add Agent Experiment**

   If other agent experiments already exist, click .

4 On the **Details** tab of the New Agent Experiment window, enter a unique, informative name for the agent experiment then select the type of experiment that you want to create.

   - If you select **Tools-based experiment**, then you must configure settings on the **Details** tab as follows.

     1 (Optional) Enter the system prompt that you want to use for all of the agent experiment's chat sessions.

     Note: If you enter a system prompt for an agent experiment, the prompt overrides any system prompt configured at the agent level and configured in the Retrieval Settings view.

     2 Enter the number of virtual CPUs that you want the experiment's container to use. Choose a value that reflects the expected workload. For example, increase CPU allocation for experiments that perform heavier processing or more frequent LLM calls.

     3 Enter the amount of RAM that you want to allocate to the experiment's container. Choose a value that supports the size and complexity of your agent's operations. Note that larger memory values might be needed for experiments that load more data or process larger prompts.

   - If you select **Template-based experiment**, then you must configure settings on the **Details** tab as follows.

     1 Select the agent template that you want to use.

     2 Select the version of the template that you want to use. The latest version is selected by default.

     3 (Optional) Enter the system prompt that you want to use for all of the agent experiment's chat sessions.

Note: If you enter a system prompt for an agent experiment, the prompt overrides any system prompt configured at the agent level and configured in the Retrieval Settings view.

4 Enter the number of virtual CPUs that you want the experiment's container to use. Choose a value that reflects the expected workload. For example, increase CPU allocation for experiments that perform heavier processing or more frequent LLM calls.

5 Enter the amount of RAM that you want to allocate to the experiment's container. Choose a value that supports the size and complexity of your agent's operations. Note that larger memory values might be needed for experiments that load more data or process larger prompts.

5 On the **LLMs** tab, select the LLM and retrieval setting that you want the agent to use.

If you are creating a template-based experiment, you must select an LLM and retrieval setting for each LLM alias that is referenced in the template. For more information about referencing LLMs by aliases in agent templates, see "Create an Agent Template" on page 76.

6 On the **Tools** tab, select which tools you want the agent to use from the list of available tools then click **+›**.

TIP You can use standard keyboard shortcuts to select multiple tools at once.

If you are creating a template-based experiment, you can select tools for each LLM alias that is referenced in the template.

7 On the **Collections** tab, select whether to enable agentic retrieval.

When agentic retrieval is enabled and you chat with the agent, the agent intelligently selects which of its available collections are relevant for retrieval, which includes applying custom metadata and tag filters. As a result, you might experience reduced processing times and costs associated with queries as opposed to when agentic retrieval is not enabled. When agentic retrieval is not enabled, the LLMs search all of their assigned collections during retrieval.

8 Select the collections from which you want the agent to retrieve information then click **+›**.

If you enabled agentic retrieval in the previous step, only collections that have a champion configuration and have a description are available for selection.

TIP You can use standard keyboard shortcuts to select multiple collections at once.

If you are creating a template-based experiment, you must select collections for each LLM alias that is referenced in the template.

9 If you are creating a template-based experiment, on the **Environment Variables** tab, enter values for each environment variable. For confidential information such as credentials, mark the environment variable as **Secret**.

10 Click **Create**.

For next steps, see “Evaluate an Agent Experiment” on page 102.

Create an Agent Experiment for an Orchestrator Agent

1. Click to navigate to the **Agents** view.
2. Double-click the name of the agent for which you want to create an experiment.
3. On the **Summary** tab, if this is the first agent experiment for the agent, click **Add Agent Experiment**

   If other agent experiments already exist, click .
4. On the **Details** tab of the New Agent Experiment window, enter a unique, informative name for the agent experiment then select the type of experiment that you want to create.
   - If you select **Code template experiment**
     1. Select the agent template that you want to use.
     2. Select the version of the template that you want to use. The latest version is selected by default.
     3. (Optional) Enter the system prompt that you want to use for all chat sessions with the orchestrator agent experiment.

        Note: If you enter a system prompt for an orchestrator agent experiment, the prompt overrides any system prompt configured at the orchestrator agent level and configured in the Retrieval Settings view.
     4. Enter the number of virtual CPUs that you want the experiment’s container to use. Select a value that reflects the expected workload. For example, increase CPU allocation for experiments that perform heavier processing or more frequent LLM calls.
     5. Enter the amount of RAM that you want to allocate to the experiment’s container. Select a value that supports the size and complexity of your agent’s operations. Note that larger memory values might be needed for experiments that load more data or process larger prompts.
   - If you select **Basic no-code experiment**, then you must configure settings on the **Details** tab as follows.

1. (Optional) Enter the system prompt that you want to use for all chat sessions with the orchestrator agent experiment.

   Note: If you enter a system prompt for an orchestrator agent experiment, the prompt overrides any system prompt configured at the orchestrator agent level and configured in the Retrieval Settings view.

2. Enter the number of virtual CPUs that you want the experiment's container to use. Choose a value that reflects the expected workload. For example, increase CPU allocation for experiments that perform heavier processing or more frequent LLM calls.

3. Enter the amount of RAM that you want to allocate to the experiment's container. Choose a value that supports the size and complexity of your agent's operations. Note that larger memory values might be needed for experiments that load more data or process larger prompts.

5. On the **LLMs** tab, select the LLM and retrieval setting that you want the agent to use.

   If you are creating a template-based experiment, you must select an LLM and retrieval setting for each LLM alias that is referenced in the template. For more information about referencing LLMs by aliases in agent templates, see "Create an Agent Template" on page 76.

6. On the **Agents** tab, select the agents that you want the orchestrator agent to be able to access.

   Note: The list of available agents includes only agents whose A2A agent card is configured.

7. If you are creating a template-based experiment, on the **Environment Variables** tab, enter values for each environment variable. For confidential information such as credentials, mark the environment variable as **Secret**.

8. Click **Create**.

Edit an Agent Experiment

1. Click to navigate to the **Agents** view.
2. Double-click the name of the agent for which you want to edit an experiment.
3. In the table, select the experiment that you want to edit then click .
4. Edit the fields on each tab as described in "Create an Agent Experiment for an Agent" on page 97 then click **Save**.

Delete an Agent Experiment

1 Click to navigate to the **Agents** view.

2 Double-click the name of the agent for which you want to edit an experiment.

3 In the table, select the experiment that you want to delete then click .

Evaluate an Agent Experiment

After you create an agent experiment, the next step is to evaluate how well it performs. Evaluating an agent experiment enables you to measure response quality, compare configurations, and identify which version delivers the most accurate and relevant results for your use case.

1 Click to navigate to the Agents view.

2 Double-click the name of the agent for which you want to evaluate an experiment.

3 Click the **User Evaluation** tab then click .

4 On the New Evaluation window, on the **Experiments** tab, select the agent experiment that you want to evaluate from the list of available experiments then click .

> **TIP** You can use standard keyboard shortcuts to select multiple experiments at once.

5 On the **Tests** tab, select the user evaluation test that you want to use to evaluate the agent experiment.

6 Click **OK**.

7 Review the following information about the evaluation in the table on the **User Evaluation** tab.

Column	Details
Timestamp	Displays the time and date on which the evaluation was initiated.
Test Case	The name of the user evaluation test

Column	Details
Experiment	The name of the agent experiment
Status	The status of the evaluation. While the evaluation is in Queued status or Running status, you can click ⊘ to cancel the evaluation. When the status is Complete, you can click ▤ to view the activity log for the evaluation.
Score	The number of evaluations passed over the number of prompts that the experiment was tested against.
Cost	The amount that the evaluation cost in USD.

8 In the bottom pane, review details about the evaluation, including the prompts that were used to evaluate the agent experiment, the agent experiment's responses to the prompts, and the result of the evaluation (whether the agent experiment passed the evaluation).

Reviewing the information in the table and the bottom pane of the **User Evaluation** tab can help you decide whether the agent experiment should be selected as the champion when completing the next steps. For next steps, see "Select a Champion Agent" on page 103.

Select a Champion Agent

After you run user evaluations on the agent experiments that you created as described in "Creating and Managing Agent Experiments" on page 97, the next step is to select a champion agent. Selecting a champion agent identifies the primary agent configuration that is used for retrieval and interaction in the Chat view.

1 Click to navigate to the **Agents** view.

2 Double-click the name of the agent for which you want to select the champion agent experiment.

3 In the table, compare the scores in the Last User Evaluation column to help determine which agent experiment should be selected as the champion.

   If necessary, compare more details about the agent experiments, such as cost, on the **User Evaluation** tab. For more information about the details on that tab, refer to the table in "Evaluate an Agent Experiment" on page 102.

4 On the **Summary** tab, use the Status column to select the champion agent experiment.

Start and Stop an Agent

Click to navigate to the Agents view. Select an agent from the list. The state of the agent is shown in the **Status** column. To start the agent, click . To stop the agent, click .

Modify an Agent

> **IMPORTANT** You cannot modify an agent that is running. You must stop the agent before you can modify it. (See "Start and Stop an Agent" on page 104.)

1 Click to navigate to the Agents view.

2 Select the agent that you want to modify.

3 Click .

4 In the Edit Agent window, you can modify properties as described in "Add an Agent" on page 96..

5 Click **OK** to save your changes.

6 (Optional) Click to restart the agent.

View Existing Agents and Their Details

To see a list of the agents that have been created for SAS Retrieval Agent Manager, click to navigate to the **Agents** view. The existing agents are listed in this view alphabetically by agent name.

To see the details that are associated with a specific agent, double-click the agent's name.

The properties that are displayed for the selected agent are described in "Edit Agent Window" on page 96..

Delete an Agent

> **IMPORTANT** You cannot delete an agent that is running. Before you can delete the agent, you must stop it. For steps to stop an agent, see "Start and Stop an Agent" on page 104.

1. Click to navigate to the **Agents** view.
2. Select the agent that you want to delete.
3. Click .
4. Click **OK** to confirm deletion.

11

Automating Workflows

About Automation

Note: This view is enabled only for authorized users.

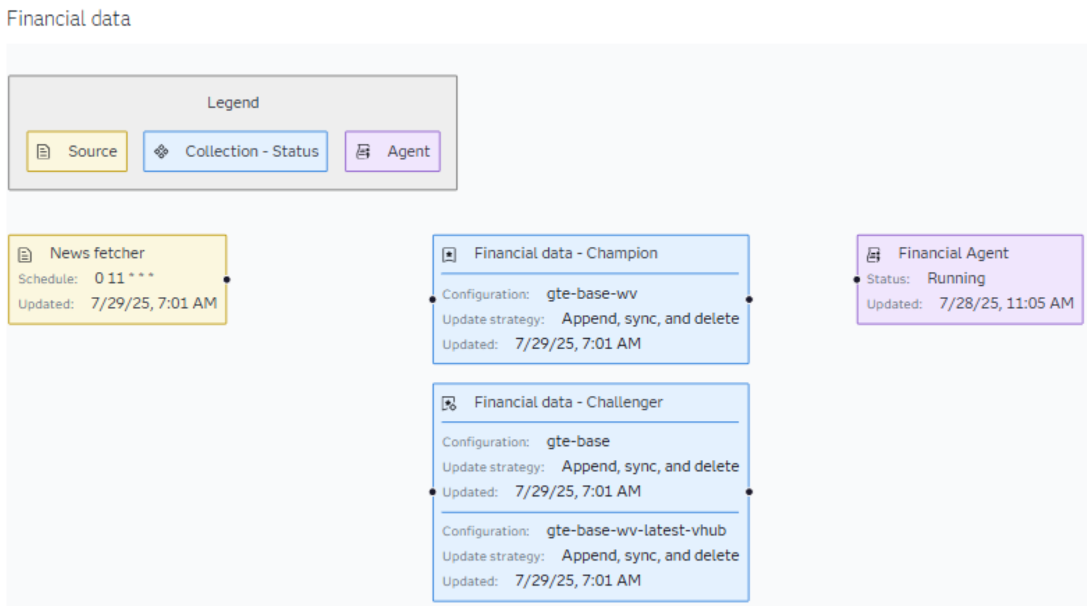
Automation enables you to connect data sources, collections, and agents within SAS Retrieval Agent Manager into scheduled workflows that run without manual intervention.

About the Automation View

The Automation view includes a list of the collections in the system and a workspace. Collections are listed in alphabetical order and are organized as **Configured pipelines** (connections exist for automation) and **Available pipelines** (no connections exist for automation).

When you select a collection, it opens in the workspace as a graphic with components that represent the data sources, configurations, and agents associated with the collection. A collection can include multiple sources and configurations. An agent is included only if it was configured to interact with the collection when the agent was created on page 96.

Here is an example of an available pipeline with no connections in the workspace:



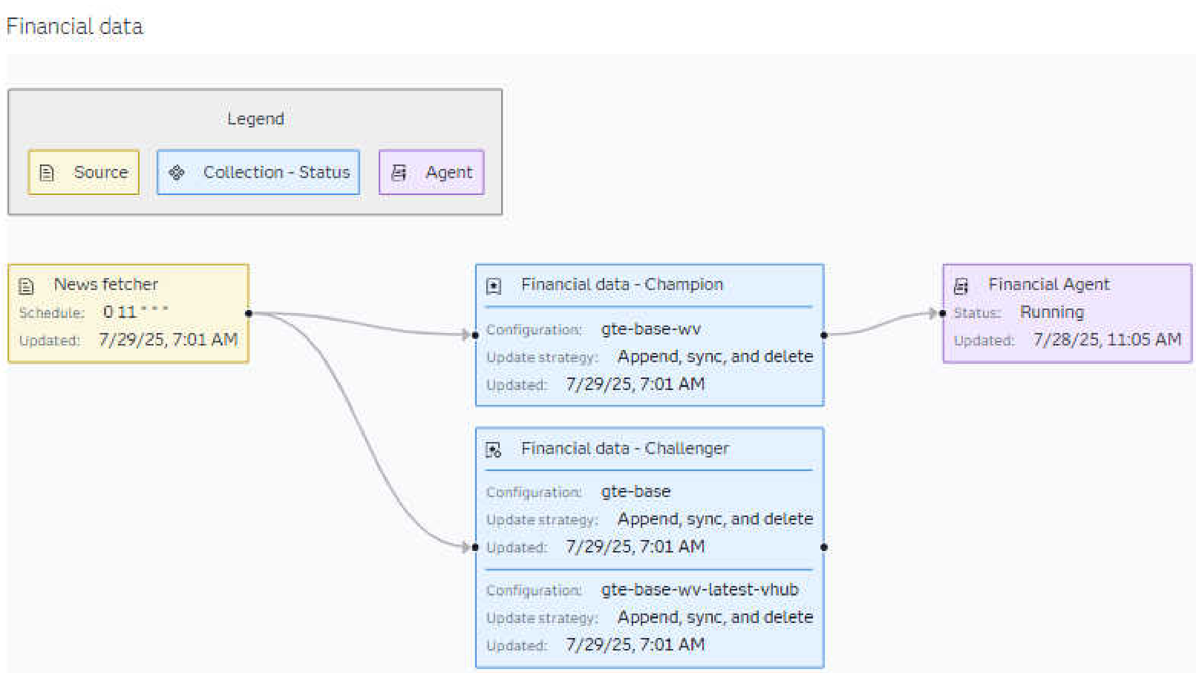
In this example, the Financial data collection includes a source, a champion configuration and two challenger configurations, and an agent. The information that is displayed about a selected pipeline is described in the following table:

Table 11.1 *Configuration Details for a Selected Pipeline*

Field		Description
Source	Schedule	The specified date, time, and frequency for when a custom file runs, in the format of a cron expression. A value of **Manual** indicates that source updates are done manually. Example: `0 0 1 * *`
	Updated	The datetime at which a source was last modified. Example: `May 30, 2035, 3:00:00 AM EST`
Collection	`The-collection-name-and-configuration-status-value`	The configuration status. A configuration can be **Under Evaluation**, a **Challenger**, or a **Champion**. Example: `Financial data - Champion`
	Update strategy	A strategy for what happens when the source documents are updated. The strategy can be **Disabled**, **Append**, **Append and sync**, or **Append, sync, and delete**. Example: `Append`
	Updated	The datetime at which a configuration was last modified. Example: `May 30, 2035, 3:00:00 AM EST`

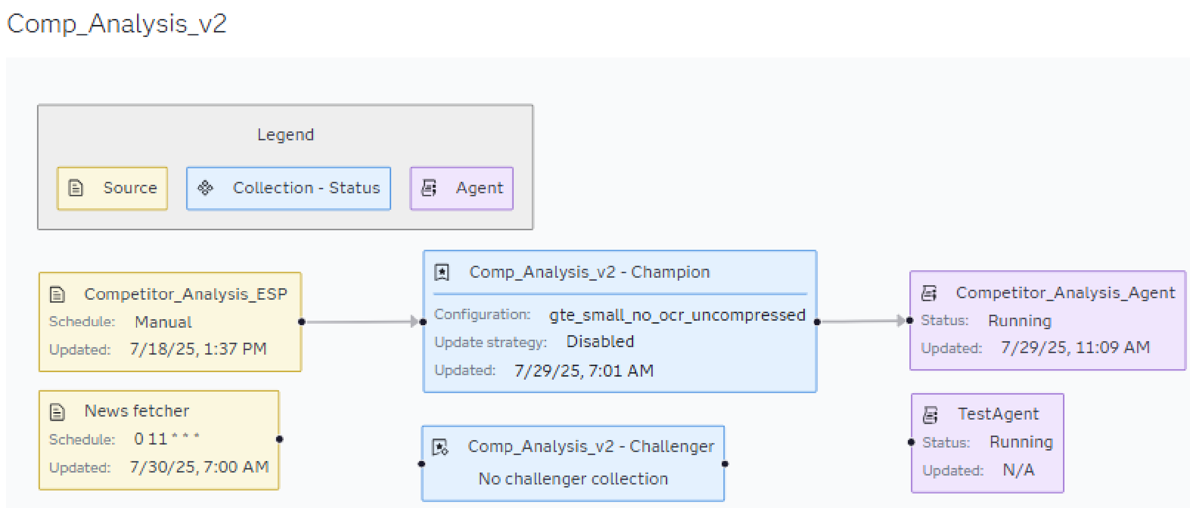
Field		Description
Agent	`Name`	The name of the agent. Example: `Financial agent`
	Status	The running status of the agent. The status can be **Running**, **Stopped**, or **Error**. Example: `Running`
	Updated	The datetime at which the agent was last initialized. Example: `May 30, 2035, 3:00:00 AM EST`

In the workspace, you can configure the connections for a pipeline by creating arrows between connection points. Here is an example of a configured pipeline with connections:



In this example, the first connections ensure that anytime the source updates, the champion and challenger configurations automatically run a vectorization job to create and save the vector embeddings for your documents to the target database. The second connection from the configuration to the agent ensures that when the vectorization job finishes, the agent is notified to initialize automatically.

Collections can have multiple sources with both manual and scheduled updates and be configured for multiple agents. Here is an example:



IMPORTANT If you connect only one source to a configuration, anytime that source updates, all sources within the collection also are re-vectorized. The synchronization strategy can be used to avoid duplicate embeddings in a configuration.

About Automation Settings

The Automation Settings window enables you to configure CPU and memory usage for automated vectorization jobs. Initially, these settings are the same settings that are used when you manually vectorize a collection on page 57. You can adjust the following settings to be specific for automated vectorization jobs.

IMPORTANT Modifying the vectorization settings for automation does not change the settings for manual vectorization.

Table 11.2 Automation Settings for Vectorization Jobs

Setting	Description
CPU	The number of CPUs that are available for vectorization. Example: `6`
Memory	The amount of memory that is available for vectorization. Example: `4Gi`
Execution provider	The execution resource. Values are **CPU** or **OpenVINO**.

Setting	Description
	Note: Although both execution resources produce the same results, vectorization jobs might run faster when you select **OpenVINO**. Example: `CPU`
OpenVINO device	(Enabled if **Execution provider** is **OpenVINO**) The device that runs the OpenVINO resource. Example: `CPU_FP32`

Configure an Automation Pipeline

1. Click to navigate to the Automation view.
2. Select a collection from the list of **Available pipelines**.
3. Connect sources to configurations and configurations to agents as desired. For each connection that you want to make, hover over a connection point. Click and hold when you see the + over the connection point. Drag an arrow to the desired connection point and release.

> **TIP** To delete an unwanted connection, click on the connection arrow and click . To revert to the previous change, click .

4. Click to save the configuration.

Modify a Configured Automation Pipeline

1. Click to navigate to the Automation view. Select a collection from the list of **Configured pipelines**.
2. You can modify the configuration in the following ways:
   - To add a connection, hover over a connection point. Click and hold when you see the + over the connection point. Drag an arrow to the desired connection point and release.

- To delete an unwanted connection, click on the connection arrow and click .

3 Click to save your changes.

Configure Settings for Automated Vectorization Jobs

> **IMPORTANT** Modifying the vectorization settings for automation does not change the settings for manual vectorization. For information about manual vectorizations, see “Run a Vectorization Job Manually” on page 57.

The settings that you can configure for automation are described in “About Automation Settings” on page 110.

1. Click to navigate to the Automation view.
2. Click .
3. In the Automated settings window, select values for **CPU**, **Memory**, and **Execution provider**. If the value that you select for **Execution provider** is **OpenVINO**, select a value for **OpenVINO device**.
4. Click **OK** to save your changes.

12

Working with the Chat Interface

About Chat

Note: Although this view is enabled for all permission groups, the features that are available to individuals with user permissions are limited.

The SAS Retrieval Agent Manager chat interface is the primary interaction point for querying the enterprise knowledge base and retrieving relevant information.

Chat View

The Chat view enables you to interact with AI-driven responses in two modes:

- On the **Collection** tab, you can run basic retrieval-augmented generation (RAG) queries by using a selected document collection and configuration.
- On the **Agent** tab, you can route basic messages through custom agents with predefined logic or behaviors.

Your ability to interact with either mode and the features that are available to you in the Chat view depend on your permissions within the system.

- Individuals with administrator permissions can interact with both collections and agents. On the **Collections** tab, administrators can see all the collections and their configurations that are in the system. They can select any of the large language models (LLMs) that are configured for a collection and select different retrieval settings. On the **Agents** tab, administrators can select an agent from the list of agents that are running in the system.

  Note: For information about starting and stopping an agent, see “Start and Stop an Agent” on page 104.

- Individuals with user permissions can interact with collections and associated agents in chat-only view if they are part of the appropriate group. A collection is listed on the **Collections** tab if the user has been given access to it by an administrator and a configuration within a collection is designated as the champion. A user cannot interact with a configuration that is under evaluation or is a challenger. A user can select an LLM if multiple models are configured for a collection and has permission to interact with those models. Users cannot select different retrieval settings.

All users can start a query based on their selections, see previous chats, rename and delete a chat, and view response details.

Chat Features

All permission groups have access to the following chat features:

Apply Tags to Queries

You can apply tags to queries by clicking , and then selecting from the list of available tags. When you apply a tag to a query, it filters the retrieval so that the query response includes only information from sources that have the selected tag. If multiple tags are applied to a query, the query response includes sources that have any or all of the selected tags.

For example, your query might be **`What follow-up care does a patient need after being discharged from the hospital?`** If you apply a *Hospital Discharge Documents* tag and an *Outpatient Follow-up Instructions* tag to the query, the retrieval pulls relevant information from sources that have the tag *Hospital Discharge Documents* and sources that have the tag *Outpatient Follow-up Instructions*, as well as sources that have both tags.

Give Feedback

By selecting or below a query response, you can indicate whether a response is good or needs improvement.

View the Query Activity Log

Select below a query response to view a color-coded activity log that lists each action that was performed when generating the response. For more information, see "View Response Details" on page 119.

View Response Metrics

You can select below a query response and then click the **Stats** tab to view the metrics from the associated query. For more information, see "View Response Details" on page 119.

View the Chat History

The chat history is displayed only for a selected configuration within a collection or for a selected agent. In the **Previous Chats** pane, you can view your previous queries and see the responses. You can rename and delete previous chats and ask follow-up queries to the chat history. You also have the option to start a new query session, which is then added to the configuration's chat history.

> **IMPORTANT** LLMs can process a certain number of tokens in their context window. If you have an unusually long chat session, you might reach the context limit. When this happens, older queries might be removed from the chat history.

Chat with Multiple Configurations

As an advanced feature, you can chat with multiple configurations across collections. Access to multiple configurations enables you to retrieve documents from all relevant collections for the query.

Start a Chat

Note: The chat interface features that are available to individuals with User permissions are limited. For more information, see “About Chat” on page 113.

1. Click to navigate to the Chat view.
2. Select whether to use a document collection or agent for a query.

To use a collection:

1. Select the **Collection** tab.
2. (Administrators only) (Optional) Under the collection that you want to query, select a configuration for the desired collection. A next to a configuration identifies the champion configuration, and a identifies a challenger configuration. A configuration that is under evaluation does not have an icon.

   Note: Individuals with User permissions see only champion configurations.

3. (Optional) Use the default **Large language model** or, if available, select a different configured model.
4. (Administrators only) (Optional) Use the default **Retrieval settings** or select a different setting.

   Note: Individuals with User permissions do not see retrieval settings.

5. In the right pane, enter a query.
6. (Optional) Click to select tags to add to the query.

   Click .

   The response to your query is displayed in the right pane. The query also is added to the **Previous Chats** pane. Click ›› to expand the **Previous Chats** pane. (For more information about previous chats, see “View and Interact with Previous Chats” on page 117.)

To use an agent:

Note: Only administrators can work with agents.

1. Select the **Agent** tab.
2. Select an agent to use.
3. In the right pane, enter a query.

4 (Optional) Click ♡ to select tags to add to the query.

5 Click ▷.

The response to your query is displayed in the right pane. The query also is added to the **Previous Chats** pane. Click ›› to expand the **Previous Chats** pane. (For more information about previous chats, see "View and Interact with Previous Chats" on page 117.)

You can continue to interact with the ongoing chat or start a new chat session for the same configuration. For more information about these tasks, see "Interact with a Previous Chat" on page 117.

View and Interact with Previous Chats

View Previous Chats

Here is how to see the chat history for a specific configuration within a collection or for a specific agent:

1 Click 🗪 to navigate to the Chat view.

2 On either the **Collection** tab or the **Agent** tab, select your query parameters as described in "Start a Chat" on page 116.

3 In the right pane, click ›› to expand the **Previous Chats** pane. (To collapse the pane, click ‹‹.)

The queries that you previously submitted for the selected parameters are listed in the **Previous Chats** pane, in order from the oldest query at the top of the list to the most recent query at the bottom. The previous queries for agents also include the datetime at which the query ran.

Interact with a Previous Chat

When you select a query from the **Previous Chats** pane, the responses for that query are displayed in the pane to the right. You can interact with previous chats in the following ways:

Give Feedback

Below a query response, you can click 👍 to indicate that a response is good or click 👎 to indicate that a response needs improvement.

Ask a Follow-up Query

Enter your query in the response pane to the right and click ▷. The response is added to the history for that chat.

> **IMPORTANT** LLMs can process a certain number of tokens in their context window. If you have an unusually long chat session, you might reach the context limit. When this happens, older queries might be removed from the chat history.

Start a New Query Session

To start a new session for the same configuration or agent, click +.

Rename a Previous Chat

Select the chat in the **Previous Chats** pane and click . In the Rename Chat Session window, enter a new **Title** and click **Confirm** to save your change.

Delete a Previous Chat

Select the chat in the **Previous Chats** pane and click . The chat is removed from the list of previous chats.

View Response Details

For both collections and agents, when you either start a new chat as described in “Start a Chat” on page 116 or interact with previous chats as described in “Interact with a Previous Chat” on page 117, you can view the metrics and the activity log that are specific to a response. In the response pane to the right for chats with either a collection or agent, click .

For both chats with a collection and chats with an agent, the left pane on the Details window displays an activity log. The activity log shows each action that was performed when generating the response. The top-level query in the activity log displays the total cost and the total duration for generating the response. When you click the top-level query, the right-pane displays the user-provided prompt and the response to that prompt. Below the top-level query is a breakdown of the subqueries, LLM calls, retrievals, and MCP tool calls that were performed to generate the response. When you click one of these entries in the activity log, the right pane displays the following information about the selected entry.

Selected Entry	Color	Details Displayed
Query	Blue	■ The prompt ■ The response to the prompt **Note:** If chatting with an agent that uses MCP tools, the response is the name of the tool that was called.
LLM Call	Green	■ The name of the LLM that was used ■ The name of the LLM provider ■ The LLM model The system prompt associated with the retrieval setting that was selected for the LLM when querying the collection ■ The prompt ■ The LLM response to the prompt **Note:** Sometimes the initial LLM response might be a reformulation of the user-provided prompt for clarity and grammatical accuracy. This LLM-generated version of the prompt might then be used in place of the original user-provided prompt for subsequent actions such as retrievals, LLM calls, and subqueries.
Retrieval	Pink	■ Tabs for each document from which chunks were retrieved

Selected Entry	Color	Details Displayed
		TIP Click a document tab to view the full document and each chunk that was retrieved from the document. ■ The names of the collections and configurations that were queried against ■ The total number of chunks in each configuration ■ The prompt that the LLM used to retrieve information ■ The number of chunks and documents that were retrieved from each configuration
The name of the MCP tool that was called	Orange	■ The name of the MCP tool server ■ The name of the MCP tool ■ The input supplied by the agent for the MCP tool call ■ The output generated by the MCP tool
The name of the agent that was called	Purple	■ The name of the agent or orchestrator agent that was called ■ The ID of the agent experiment that was used ■ The prompt that the agent used to retrieve information ■ The response generated by the agent

For chats with a collection, the **Stats** tab in the Details window displays two ring charts. The **Response Duration** chart shows the **Total Duration** in seconds for the pre-processing, retrieval, and chat stages of a query. You can hover over the chart to see the duration for each stage of the chat. The **Response Cost** chart shows the **Total Cost** in USD for the prompt and completion. You can hover over the chart to see the cost for each stage. To see where the LLM is getting the information for the response, you can select the document file in the left pane to see the most relevant chunks of data that were returned from the query.

For chats with an agent, the **Stats** tab in the Details window displays metrics for the number of **Intermediate Queries**, the **Total Duration** of the query in seconds, the **Total Prompt Cost** in USD, the **Total Completion Cost** in USD, and the **Total Cost** in USD.

Click **Close** to exit a details window.

13

Supported File Types

Supported File Types

The following table lists the file types that are supported with SAS Retrieval Agent Manager.

Table 13.1 *Supported File Types*

File Type	File Extension
Email message	.eml
Epub	.epub
Excel	.csv, .tsv, .xls, .xlsx
HTML	.htm, .html
Images	.jpeg, .jpg, .png
JSON	.json
Markdown	.md
Message file	.msg
ODS spreadsheet	.ods

File Type	File Extension
OpenDocument text format	.odt
Org text files	.org
PDF	.pdf
Powerpoint	.pptx **Note:** The .ppt file extension is not supported.
Restructured text file	.rst
Rich text file	.rtf
Text file	.txt
SAS data set file	.sas7bdat
Source code	.c, .cc, .cobol, .cpp, .cs, .cxx, .go, .h, .hpp, .java, .js, .kt, .kts, .lua, .perl, .php, .proto, .py, .rb,
Word	.docx **Note:** The .doc file extension is not supported.
XML	.xml